United States Patent	12414069
Kind Code	B2
Date of Patent	September 09, 2025
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Network search method for SA 5G service auto-provisioning and related apparatus

Abstract

A network search method for standalone (SA) fifth generation (5G) service auto-provisioning includes a user equipment sending a registration request to a network side device to request to register the user equipment with a 5G network. A networking mode of the 5G network is SA networking. The user equipment receives a registration reject response from the network side device when no available 5G subscription data of the user equipment is obtained through query, where the registration reject response notifies the user equipment that the registration with the 5G network fails. The user equipment receives a notification short message from the network side device after the network side device provisions a 5G service for the user equipment; and the user equipment sends the registration request to the network side device again in response to the notification short message.

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Appl. No.:	**17/921442**
Filed (or PCT Filed):	**April 26, 2021**
PCT No.:	**PCT/CN2021/090038**
PCT Pub. No.:	**WO2021/218936**
PCT Pub. Date:	November 04, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230164727 A1	May. 25, 2023

Foreign Application Priority Data

CN	202010339740.3	Apr. 26, 2020

Publication Classification

Int. Cl.: **H04W60/00** (20090101)

U.S. Cl.:

CPC **H04W60/00** (20130101);

Field of Classification Search

CPC: H04W (60/00); H04W (48/18); H04W (4/14); H04W (4/50); H04W (60/04); H04W (48/16)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2005/0020259	12/2004	Herrero	455/433	H04W 8/12
2021/0258857	12/2020	Won	N/A	H04W 48/18
2023/0189187	12/2022	Velev	455/435.1	H04W 48/16

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2019034610	12/2018	WO	H04W 60/04

OTHER PUBLICATIONS

3GPP TS 24.501 V0.1.0, Nov. 2017, “3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3 (Release 15),” 16 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a U.S. National Stage of International Patent Application No. PCT/CN2021/090038 filed on Apr. 26, 2021, which claims priority to Chinese Patent Application No. 202010339740.3 filed on Apr. 26, 2020. Both of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) This application relates to the field of communication technologies, and in particular, to a network search method for SA 5G service auto-provisioning and a related apparatus.

BACKGROUND

(3) With advent of a fifth-generation mobile communication technology (5th generation mobile networks, 5G) era, a 5G mobile network can meet people's communication requirements, and can also support connections of massive interact of things devices, to provide mobile communication services for thousands of industries. Deployment modes of 5G networking mainly include the following two modes: non-standalone (non-standalone, NSA) networking and standalone (standalone, SA) networking. In non-standalone networking, existing 4G infrastructure is used to deploy a 5G network. The standalone networking mode features a mature technology, large-scale coverage, and the like. In the standalone networking mode, a new 5G network is established, and the 5G network includes new base stations, backhaul links, and a core network. This mode can bring all 5G network features and functions. Compared with NSA networking, SA networking has a better performance advantage. However, standalone networking requires continuous investment in new devices and has higher costs than those of non-standalone networking. In an initial phase, an operator will focus on NSA. However, in the near future, SA will gradually replace NSA and become a mainstream in the market. When a mobile phone intends to register with a SA 5G network, a SA 5G service needs to be provisioned in advance. For a SA 5G network registration request from a mobile phone for which a SA 5G service is not provisioned, different operators process the request in different ways.

(4) Currently, user equipment for which a SA 5G service is not provisioned cannot quickly register with a SA 5G network, and user experience is poor.

SUMMARY

(5) Embodiments of this application provide a network search method for SA 5G service auto-provisioning and a related apparatus, so that user equipment for which a SA 5G service is not provisioned can quickly register with a SA 5G network, and user experience is effectively improved.

(6) According to a first aspect, this application provides a network search system for SA 5G service auto-provisioning, where the system includes user equipment and a network side device, where the user equipment is configured to send a registration request to a network side device, where the registration request is used to request to register the user equipment with a, 5G network, and a networking mode of the 5G network is standalone networking SA; the network side device is configured to: in response to the registration request, send a registration reject response to the user equipment when no available 5G subscription data corresponding to the user equipment is obtained through query, and provision a 5G service for the user equipment, where the registration reject response is used to notify the user equipment that the registration with the 5G network fails; the network side device is further configured to send a notification short message to the user equipment after the 5G service is successfully provisioned for the user equipment, where the notification short message is used to notify the user equipment that the 5G service is successfully provisioned; and the user equipment is further configured to: in response to the received notification short message,

send a registration request to the network side device again.

(7) In an implementation, that the user equipment is configured to send a registration request to a network side device includes: the user equipment is specifically configured to send the registration request to the network side device for the first time.

(8) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment has no network service in an NR cell, send the registration request to the network side device for the first time.

(9) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment leaves an LTE cell and enters a new radio NR cell, send the registration request to the network side device for the first time.

(10) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment is powered on in an NR cell, send the registration request to the network side device for the first time.

(11) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when an airplane mode of the user equipment is disabled in an NR cell, send the registration request to the network side device for the first time.

(12) In an implementation, the user equipment supports the 5G network.

(13) In an implementation, after the network side device provisions the 5G service for the user equipment, the network side device stores the available 5G subscription data corresponding to the user equipment.

(14) In an implementation, the network side device is further configured to: in response to the registration request sent by the user equipment again, send a registration accept response to the user equipment after the available 5G subscription data corresponding to the user equipment is obtained through query, where the registration accept response is used to notify the user equipment that the registration with the 5G network succeeds.

(15) In an implementation, that the user equipment is further configured to: in response to the received notification short message, send the registration request to the network side device again includes: the user equipment is further specifically configured to: when the user equipment detects that the notification short message includes a preset keyword, send the registration request to the network side device again.

(16) According to a second aspect, this application provides a network search method for SA 5G service auto-provisioning. The method includes: user equipment sends a registration request to a network side device, where the registration request is used to request to register the user equipment with a 5G network, and a networking mode of the 5G network is standalone networking SA; the user equipment receives a registration reject response sent by the network side device when no available 5G subscription data corresponding to the user equipment is obtained through query, where the registration reject response is used to notify the user equipment that the registration with the 5G network fails; the user equipment receives a notification short message sent by the network side device after the network side device provisions a 5G service for the user equipment, where the notification short message is used to notify the user equipment that the 5G service is successfully provisioned; and the user equipment sends, in response to the notification short message, the registration request to the network side device again.

(17) In this embodiment of this application, user equipment for which a 5G service is not provisioned sends a 5G network registration request to the network side device, and the network side device queries, in response to the registration request, 5G subscription data corresponding to the user equipment. When no available 5G subscription data of the user equipment is obtained

through query, the network side device sends a registration reject response to the user equipment, and provisions the 5G service for the user equipment. Then, the network side device sends a notification short message to the user equipment after the 5G service is successfully provisioned, to notify the user equipment that the 5G service is successfully provisioned. When determining, based on the notification short message, that the 5G service is successfully provisioned, the user equipment initiates a 5G network registration request again. In this way, the user equipment can quickly register with the 5G network, and user experience is effectively improved.

(18) In an implementation, that the user equipment sends the registration request to the network side device includes: the user equipment sends the registration request to the network side device for the first time.

(19) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment has no network service in an NR cell, sending the registration request to the network side device for the first time.

(20) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment leaves an LTE cell and enters a new radio NR cell, sending the registration request to the network side device for the first time.

(21) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment is powered on in an NR cell, sending the registration request to the network side device for the first time.

(22) In an implementation, that the user equipment sends the registration request to the network side device includes: when an airplane mode of the user equipment is disabled in an NR cell, sending the registration request to the network side device for the first time.

(23) In an implementation, the user equipment supports the 5G network.

(24) In an implementation, after the network side device provisions the 5G service for the user equipment, the network side device stores the available 5G subscription data corresponding to the user equipment.

(25) In an implementation, after the user equipment sends the registration request to the network side device again, the method further includes: the user equipment receives the registration accept response sent by the network side device after the available 5G subscription data corresponding to the user equipment is obtained through query, where the registration accept response is used to notify the user equipment that the registration with the 5G network succeeds.

(26) In an implementation, that the user equipment sends, in response to the notification short message, the registration request to the network side device again includes: when the user equipment detects that the notification short message includes a preset keyword, sending the registration request to the network side device again.

(27) According to a third aspect, this application further provides a network search system for SA 5G service auto-provisioning, where the system includes user equipment, a first network device, and a maintenance system of an operator, where the user equipment is configured to send a registration request to the first network device, where the registration request is used to request to register the user equipment with a 5G network, and a networking mode of the 5G network is standalone networking SA; the first network device is configured to: in response to the registration request, send a registration reject response to the user equipment when no available 5G subscription data corresponding to the user equipment is obtained through query, and sends a provisioning request to the maintenance system, where the registration reject response is used to notify the user equipment that the registration with the 5G network fails; the maintenance system is configured to: in response to the provisioning request, provision a 5G service for the user equipment, and sends a notification short message to the user equipment after the 5G service is successfully provisioned, where the notification short message is used to notify the user equipment that the 5G service is successfully provisioned; and the user equipment is further configured to: in response to the notification short message, send a registration request to the first network device

again.
(28) According to a fourth aspect, this application provides a network search system for SA 5G service auto-provisioning, where the system includes user equipment and a network side device, where the user equipment is configured to send a registration request to a network side device, where the registration request is used to request to register the user equipment with a 5G network, and a networking mode of the 5G network is standalone networking SA; the network side device is configured to: in response to the registration request, send a registration reject response to the user equipment when no available 5G subscription data corresponding to the user equipment is obtained through query, and provision a 5G service for the user equipment, where the registration reject response is used to notify the user equipment that the registration with the 5G network fails, the registration reject response includes reject cause information, and the reject cause information is used to indicate a cause of the registration failure; the network side device is further configured to send a detach request to the user equipment after the network side device provisions the 5G service for the user equipment, where the detach request is used to trigger the user equipment to re-register with a 4G network; and the user equipment is further configured to: in response to the reject cause information and the detach request, send a registration request to the network side device again.
(29) In an implementation, the reject cause information is a reject cause value #111.
(30) In an implementation, the reject cause information is a reject cause value #127.
(31) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to send the registration request to the network side device for the first time.
(32) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment has no network service in an NR cell, send the registration request to the network side device for the first time.
(33) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment leaves an LIE cell and enters a new radio NR cell, send the registration request to the network side device for the first time.
(34) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment is powered on in an NR cell, send the registration request to the network side device for the first time.
(35) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when an airplane mode of the user equipment is disabled in an NR cell, send the registration request to the network side device for the first time.
(36) In an implementation, the user equipment supports the 5G network.
(37) In an implementation, after the network side device provisions the 5G service for the user equipment, the network side device stores the available 5G subscription data corresponding to the user equipment.
(38) In an implementation, the network side device is further configured to: in response to the registration request sent by the user equipment again, send a registration accept response to the user equipment after the available 5G subscription data corresponding to the user equipment is obtained through query, where the registration accept response is used to notify the user equipment that the registration with the 5G network succeeds.
(39) According to a fifth aspect, this application provides a network search method for SA 5G service auto-provisioning. The method includes: user equipment sends a registration request to a network side device, where the registration request is used to request to register the user equipment with a 5G network, and a networking mode of the 5G network is standalone networking the user

equipment receives a registration reject response sent by the network side device when no available 5G subscription data corresponding to the user equipment is obtained through query, where the registration reject response is used to notify the user equipment that the registration with the 5G network fails, the registration reject response includes reject cause information, and the reject cause information is used to indicate a cause of the registration failure the network side device is configured to provision a 5G service for the user equipment when no available 5G subscription data corresponding to the user equipment is obtained through query the user equipment receives a detach request that is sent by the network side device after the network side device provisions the 5G service for the user equipment, where the detach request is used to trigger the user equipment to re-register with a 4G network; and the user equipment sends, in response to the reject cause information and the detach request, a registration request to the network side device again.
(40) In this embodiment of this application, user equipment for which a 5G service is not provisioned sends a 5G network registration request to the network side device, and the network side device queries, in response to the registration request, 5G subscription data corresponding to the user equipment. When no available 5G subscription data of the user equipment is obtained through query, the network side device sends a registration reject response to the user equipment, and provisions the 5G service for the user equipment. The network side device adds the reject cause information to the registration reject response, and sends the detach request to the user equipment after the network side device provisions the 5G service for the user equipment. When receiving the detach request after receiving the reject cause information, the user equipment initiates a 5G network registration request again. In this way, user equipment for which a SA 5G service is not provisioned can also quickly register with the 5G network, and user experience is effectively improved.
(41) In an implementation, the reject cause information is a reject cause value #111.
(42) In an implementation, the reject cause information is a reject cause value #127.
(43) In an implementation, a timer is numbered as T**3502**.
(44) In an implementation, that the user equipment sends the registration request to the network side device includes: the user equipment sends the registration request to the network side device for the first time.
(45) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment has no network service in an NR cell, sending the registration request to the network side device for the first time.
(46) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment leaves an LTE cell and enters a new radio NR cell, sending the registration request to the network side device for the first time.
(47) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment is powered on in an NR cell, sending the registration request to the network side device for the first time.
(48) In an implementation, that the user equipment sends the registration request to the network side device includes: when an airplane mode of the user equipment is disabled in an NR cell, sending the registration request to the network side device for the first time.
(49) In an implementation, the user equipment supports the 5G network.
(50) In an implementation, after the user equipment sends the registration request to the network side device again, the method further includes: the user equipment receives the registration accept response sent by the network side device after the available 5G subscription data corresponding to the user equipment is obtained through query, where the registration accept response is used to notify the user equipment that the registration with the 5G network succeeds.
(51) According to a sixth aspect, this application provides a network search system for SA 5G service auto-provisioning, where the system includes user equipment and a network side device, where the user equipment is configured to send a registration request to a network side device,

where the registration request is used to request to register the user equipment with a 5G network, and a networking mode of the 5G network is standalone networking SA; the network side device is configured to: in response to the registration request, send a registration reject response to the user equipment when no available 5G subscription data corresponding to the user equipment is obtained through query, and provision a 5G service for the user equipment, where the registration reject response is used to notify the user equipment that the registration with the 5G network tails, the registration reject response includes reject cause information, and the reject cause information is used to indicate a cause of the registration failure; the user equipment is further configured to: in response to the reject cause information, start a timer; and the user equipment is further configured to: in response to expiration of the timer, send a registration request to the network side device again.
(52) In an implementation, the reject cause information is a reject cause value #111.
(53) In an implementation, the reject cause information is a reject cause value #127.
(54) In an implementation, the timer is numbered as T**3502**.
(55) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to send the registration request to the network side device for the first time.
(56) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment has no network service in an NR cell, send the registration request to the network side device for the first time.
(57) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment leaves an LIE cell and enters a new radio NR cell, send the registration request to the network side device for the first time.
(58) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when the user equipment is powered on in an NR cell, send the registration request to the network side device for the first time.
(59) In an implementation, that the user equipment is configured to send the registration request to the network side device includes: the user equipment is specifically configured to: when an airplane mode of the user equipment is disabled in an NR cell, send the registration request to the network side device for the first time.
(60) In an implementation, the user equipment supports the 5G network.
(61) In an implementation, after the network side device provisions the 5G service for the user equipment, the network side device stores the available 5G subscription data corresponding to the user equipment.
(62) In an implementation, the network side device is further configured to: in response to the registration request sent by the user equipment again, send a registration accept response to the user equipment after the available 5G subscription data corresponding to the user equipment is obtained through query where the registration accept response is used to notify the user equipment that the registration with the 5G network succeeds.
(63) According to a seventh aspect, this application provides a network search method for SA 5G service auto-provisioning. The method includes: user equipment sends a registration request to a network side device, where the registration request is used to request to register the user equipment with a 5G network, and a networking mode of the 5G network is standalone networking SA; the user equipment receives a registration reject response sent by the network side device when no available 5G subscription data corresponding to the user equipment is obtained through query, where the registration reject response is used to notify the user equipment that the registration with the 5G network fails, the registration reject response includes reject cause information, and the

reject cause information is used to indicate a cause of the registration failure; the network side device is configured to provision a 5G service for the user equipment when no available 5G subscription data corresponding to the user equipment is obtained through query; the user equipment starts, in response to the reject cause information, a timer; and the user equipment sends, in response to expiration of the timer, a registration request to the network side device again.
(64) In this embodiment of this application, user equipment for which a 5G service is not provisioned sends a 5G network registration request to the network side device, and the network side device queries, in response to the registration request, 5G subscription data corresponding to the user equipment. When no available 5G subscription data of the user equipment is obtained through query, the network side device sends a registration reject response to the user equipment, and provisions the 5G service for the user equipment. The network side device adds the reject cause information to the registration reject response, and the reject cause information may trigger the user equipment to start the timer. In addition, when determining that the timer expires, the user equipment initiates a 5G network registration request again. The network side device has provisioned the 5G service for the user equipment before the timer expires. Therefore, the user equipment can quickly register with the 5G network, and user experience is effectively improved.
(65) In an implementation, the reject cause information is a reject cause value #111.
(66) In an implementation, the reject cause information is a reject cause value #127.
(67) In an implementation, the timer is numbered as T**3502**.
(68) In an implementation, that the user equipment sends the registration request to the network side device includes: the user equipment sends the registration request to the network side device for the first time.
(69) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment has no network service in an NR cell, sending the registration request to the network side device for the first time.
(70) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment leaves an LTE cell and enters a new radio NR cell, sending the registration request to the network side device for the first time.
(71) In an implementation, that the user equipment sends the registration request to the network side device includes: when the user equipment is powered on in an NR cell, sending the registration request to the network side device for the first time.
(72) In an implementation, that the user equipment sends the registration request to the network side device includes: when an airplane mode of the user equipment is disabled in an NR cell, sending the registration request to the network side device for the first time.
(73) In an implementation, the user equipment supports the 5G network.
(74) In an implementation, after the user equipment sends the registration request to the network side device again, the method further includes: the user equipment receives the registration accept response sent by the network side device after the available 5G subscription data corresponding to the user equipment is obtained through query, where the registration accept response is used to notify the user equipment that the registration with the 5G network succeeds.
(75) According to an eighth aspect, user equipment is provided, including a communication interface, a memory, and a processor. The communication interface and the memory are coupled to the processor, the memory is configured to store computer program code, the computer program code includes computer instructions. When the processor reads the computer instructions from the memory, the user equipment is enabled to perform any possible implementation of the second aspect, any possible implementation of the fifth aspect, or any possible implementation of the seventh aspect.
(76) According to a ninth aspect, a chip is provided, including a memory and a processor. The memory is coupled to the processor, the processor includes a modem processor, the memory is configured to store computer program code, and the computer program code includes computer

instructions. When the processor reads the computer instructions from the memory, the chip is enabled to perform any possible implementation of the second aspect, any possible implementation of the fifth aspect, or any possible implementation of the seventh aspect.
(77) According to a tenth aspect, a computer-readable storage medium is provided, including instructions, where when the instructions are run on user equipment, the user equipment is enabled to perform any possible implementation of the second aspect, any possible implementation of the fifth aspect, or any possible implementation of the seventh aspect.
(78) According to an eleventh aspect, a computer product is provided. When the computer program product runs on a computer, the computer is enabled to perform any possible implementation of the second aspect, any possible implementation of the fifth aspect, or any possible implementation of the seventh aspect.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a schematic diagram of a system architecture of a communication system according to an embodiment of this application;
(2) FIG. **2** to FIG. **8**A and FIG. **8**B are examples of user interfaces according to an embodiment of this application;
(3) FIG. **9** is a schematic flowchart of a network search method for SA 5G service auto-provisioning according to an embodiment of this application;
(4) FIG. **10**A and FIG. **10**B are a schematic flowchart of another network search method for SA 5G service auto-provisioning according to an embodiment of this application;
(5) FIG. **11** to FIG. **14** are examples of user interfaces of short messages according to an embodiment of this application;
(6) FIG. **15**A and FIG. **15**B are a schematic flowchart of still another network search method for SA 5G service auto-provisioning according to an embodiment of this application;
(7) FIG. **16**A, FIG. **16**B and FIG. **16**C are a schematic flowchart of yet another network search method for SA 5G service auto-provisioning according to an embodiment of this application;
(8) FIG. **17** is a schematic flowchart of still yet another network search method for SA 5G service auto-provisioning according to an embodiment of this application;
(9) FIG. **18**A and FIG. **18**B are a schematic flowchart of a further network search method for SA 5G service auto-provisioning according to an embodiment of this application;
(10) FIG. **19**A and FIG. **19**B to FIG. **22**A and FIG. **22**B are examples of user interfaces according to an embodiment of this application;
(11) FIG. **23** is a schematic diagram of a structure of user equipment according to an embodiment of this application;
(12) FIG. **24** is a block diagram of a software structure of user equipment according to an embodiment of this application; and
(13) FIG. **25** is a block diagram of a software structure of a network device according to an embodiment of this application.
DESCRIPTION OF EMBODIMENTS
(14) The following clearly describes the technical solutions in embodiments of this application in detail with reference to the accompanying drawings. In descriptions of embodiments of this application, unless otherwise specified, "/" indicates "or". For example, A/B may indicate A or B. The term "and/or" in this specification merely describes an association relationship for describing associated objects, and indicates that three relationships may exist. For example, A and/or B may indicate the Mowing three cases: Only A exists, both A and B exist, and only B exists. In addition, in the descriptions of embodiments of this application, "a plurality of" means two or more.

(15) The following terms "first" and "second" are merely intended for description, and shall not be understood as an indication or implication of relative importance or implicit indication of a quantity of indicated technical features. Therefore, a feature limited by "first" or "second" may explicitly or implicitly include one or more features. In the descriptions of embodiments of this application, unless otherwise specified, "a plurality of" means two or more.
(16) FIG. **1** shows a network architecture of a communication system **100** according to an embodiment of this application. As shown in FIG. **1**, the communication system **100** may include: user equipment (user equipment, UE) **101**, a (Radio) access network (access network, AN) device **102**, a user plane function (user plane function, UM entity **103**, a data network (data network, DN) device **104** and a control plane functional entity. The control plane functional entity may include: an AMF (access and mobility management function, access and mobility management function) entity **105**, an SMF (session management function, session management function) entity **106**, an authentication service function (authentication server function, AUSF) entity **107**, an NSSF **108**, an NET **109**, an NF repository function (NF repository function, NRF) entity **110**, a policy control function (policy control function, PCF) entity **111**, a unified data management (unified data management, UDM) entity **112**, and an application function (application function, AF) entity **113**.
(17) The UE **101** in this embodiment of this application may be a wireless terminal device with a wireless connection function. The UE **101** may be distributed in the entire communication system **100**, and may be stationary or mobile. The UE **101** may communicate with one or more core networks by using the (R)AN **102**. For example, the UE **101** may be a mobile terminal device such as a mobile phone, a computer, a tablet computer, a personal communication service (personal communication service. PCS) phone, or a personal digital assistant (personal digital assistant, PDA); or the UE **101** may be a mobile station (mobile station), a mobile unit (mobile unit), an M2M terminal, a wireless unit, a remote unit, a terminal agent, a mobile client, or the like; or the UE **101** may be an internet of things terminal device or an internet of vehicles terminal device. This is not limited in this embodiment of this application.
(18) The (R)AN **102** may be a 5G base station, an NR base station, or the like. The UE **101** may access the communication system **100** by using the (R)AN **102**. Alternatively, the (R)AN **102** may be a network including a plurality of (R)AN **102** nodes, and may implement functions such as a wireless physical layer function, a resource scheduling function, and a radio resource management function. The (R)AN **102** is connected to the UPF **103** through a user plane interface N3, and may be configured to transmit data of the UE **101**. The AN establishes a control plane signaling connection to the AMF **105** through a control plane interface N2, and may be configured to implement functions such as radio access bearer control.
(19) The UPF **103** may be used for functions such as packet routing and forwarding, quality of service (Quality of Service, QoS) processing of a user plane, uplink traffic verification, transport-level packet marking in an uplink and a downlink, downlink data packet buffering, and downlink data notification triggering. The UPF **103** is connected to the SMF **106** through an interface N4, and the UPF **103** is connected to the DN **104** through an interface N6. The UPF **103** is a session point for interconnection between an external protocol data unit (protocol data unit, PDU) and a data network.
(20) The AMF **105** may be used for functions such as authentication on the UE **101**, SMF **106** selection, connection management, reachability management, and mobility management. The AMF **105** may be further configured to provide transmission for an SM message between the UE **101** and the SMF **106**. The AMF **105** may serve as an anchor for a signaling connection between N1 and N2, to maintain and manage status information of the UE **101**.
(21) The SMF **106** may be configured to be responsible for all control plane functions of session management of the terminal device, including UPF **103** selection, internet protocol (internet protocol, IP) address allocation, QoS attribute management of a session, and the like. The SMF **106** is connected to the UPF **103** through the interface N4. The SMF **106** may control insertion and

removal of an uplink classifier, and may further control insertion and removal of the UPF **103** that supports a branching point.
(22) The AUSF **107** is configured to interact with the UDM **112** to obtain user information, and perform functions related to authentication and authorization, for example, generating an intermediate key.
(23) The NSSF **108** is configured to: select a network slice for the UE **101**, and send signaling to a corresponding network slice based on single network slice selection assistance information (single network slice selection assistance information, S-NSSAI).
(24) The NEF **109** is configured to: expose various services and capabilities (including content exposure or exposure to a third party, and the like) provided by 3GPP network functions, perform verification and authorization, and assist in limiting an application function.
(25) The NRF **110** is configured to: receive an NF discovery request from an NF instance, and provide the discovered NF instance information for the NF instance, to maintain the NF instance and an NF configuration file.
(26) The PCF **111** is configured to: formulate a policy, provide a policy control service, obtain policy-related subscription data, and the like.
(27) The UDM **112** is configured to: store user subscription data corresponding to the UE **101**, perform user identification processing, support functions such as service/session continuity, access authorization, registration and mobility management, subscription management, and user management, and the like.
(28) The AF **113** is configured to: interact with a 3GPP core network to provide a service. The AF **113** may access a network exposure function, and control policy framework interaction.
(29) It should be noted that physically, each entity shown in FIG. **1** may be a single device, or two or more entities may be integrated into a same physical device. This is not limited in this embodiment of this application.
(30) In this embodiment of this application, the communication system **100** may be a 5G communication system, an NR system, a future evolved communication system, or the like.
(31) In this embodiment of this application, the core network entity may alternatively be referred to as a core network device or a network device.
(32) As shown in FIG. **1**, deployment of a 5G network mainly needs two parts: a radio access network (radio access network. RAN) and a core network (core network). The radio access network mainly consists of base stations, and provides a radio access function for a user. The core network mainly provides an internet access service and a corresponding management function for the user. Investment in deploying a new network is huge. Therefore, 3GPP (3rd generation partnership project, a standardization organization) defines the following two 5G network deployment modes: SA and NSA. In the SA mode, a new existing network is established, including a new base station, a new backhaul link, and a new core network. In the NSA mode, existing 4G infrastructure is used to deploy a 5G network.
(33) In the NSA mode, a control plane is independently deployed on the 4G network, and a user plane is co-deployed on the 5G network and the 4G network, or a user plane is independently deployed on the 5G network. In the NSA mode, an advantage of a high bandwidth of the 5G network may be introduced, popularization of the 5G network is accelerated, and device replacement costs are reduced in a short term, in the SA mode, the control plane and the user plane are independently deployed on the 5G network. An independent networking based on a 5G new radio technology may work without depending on the 4G network. In the SA mode, a new network element and a new interface are introduced, and at the same time, new technologies such as network virtualization and software-defined networking are to be used on a large scale. In an initial phase, an operator will focus on NSA networking. However, in the future, SA networking will gradually replace NSA networking and become a mainstream in the market.
(34) For ease of description, 5G in a SA deployment mode may be referred to as SA 5G for short in

this specification, or a process of provisioning (subscribing to) a SA 5G service may be referred to as SA 5G service provisioning for short.
(35) Generally, when a mobile phone is in a state in which no network service is available, the mobile phone searches for a network and initiates initial registration. For example, the mobile phone is powered on, an airplane mode of the mobile phone is disabled, or the mobile phone is disconnected from a network. In a default setting of the mobile phone, the mobile phone preferably searches for a highest-standard network. For example, when the mobile phone supports 3G, 4G, and 5G networks, the mobile phone first searches for a 5G network. A user may also select, in settings of the mobile phone, a network mode searched first.
(36) For ease of understanding, the following describes some example graphical user interfaces implemented on user equipment **200** provided in an embodiment of this application by using examples in which an airplane mode is disabled and the user equipment **200** is restarted and powered on. The user equipment **200** may be the UE **101** in the communication system **200** shown in FIG. **1**.
(37) FIG. **2** shows an example of a user interface **10** that is on the user equipment **200** and that is used to display applications installed on the user equipment **200**.
(38) The user interface **10** may include: a status bar **201**, a calendar indicator **202**, a weather indicator **203**, a bottom tray **204** including commonly used application icons, a navigation bar **205**, and another application icon.
(39) The status bar **201** may include: an airplane mode indicator **201**A, a battery status indicator **201**B, and a time indicator **201**C.
(40) It can be learned from the status bar shown in FIG. **2** that the user equipment **200** is currently in an airplane mode.
(41) The calendar indicator **202** may be used to indicate current time, such as a date, a day of a week, hour-minute information, and the like.
(42) The weather indicator **203** may be used to indicate a weather type, for example, cloudy to clear or light rain, and may be further used to indicate information such as a temperature.
(43) The bottom tray **204** including commonly used application icons may display: a “Phone” icon **204**A, a “Contacts” icon **204**B, a “Messages” icon **204**C, and a “Camera” icon **204**D.
(44) The navigation bar **205** may include system navigation buttons such as a “Back” button **205**A, a “Home screen” button **205**B, and a “Multitask” button **205**C. When it is detected that a user taps the “Back” button **205**A, the user equipment **200** may display a previous page of a current page. When it is detected that the user taps the “Home screen” button **205**B, the user equipment **200** may display a home screen. When it is detected that the user taps the “Multitask” button **205**C, the user equipment **200** may display a task recently opened by the user. Alternatively, the navigation buttons may be named in another way. This is not limited in this application, Each navigation button in the navigation bar **205** is not limited to a virtual button, and may alternatively be implemented as a physical button.
(45) Other application icons may be, for example, a “Transfer” icon **206**, a “Gallery” icon **207**, a “Music” icon **208**, an “Application” icon **209**, an “Email” icon **210**, a “Cloud share” icon **211**, a “Notepad” icon **212**, and a “Settings” icon **213**. The user interface **10** may further include a page indicator **214**. The other application icons may be distributed on a plurality of pages, and the page indicator **216** may be used to indicate a specific page on Which an application is currently browsed by the user. The user may slide leftward or rightward in an area including the other application icons, to browse an application icon on another page.
(46) In some embodiments, the user interface **10** shown in FIG. **2** as an example may be a home screen (Home screen). It may be understood that FIG. **2** merely shows an example of a user interface on the user equipment **200**, and should not constitute a limitation on this embodiment of this application.
(47) For example, as shown in FIG. **3**A and FIG. **3**B, the user slides a finger downward from a top

of a display and then releases the finger. The user equipment **200** detects the user operation, and in response to the user operation, the user equipment **200** displays a user interface **11** of a drop-down bar.
(48) The user interface **11** may include: a status bar **301**, a navigation bar **302**, time **303**, a settings icon **304**, a quick settings panel **305**, a brightness control **306**, and one or more system notification messages **307**.
(49) The settings icon **304** may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** may display a setting interface of a drop-down bar.
(50) The quick settings panel **305** may display: a “Wi-Fi” icon **305**A, a “Bluetooth” icon **305**B, a “Flashlight” icon **305**C, a “Ringing” icon **305**D, an “Auto-rotate” icon **305**E, a “Message center” icon **305**F, an “Airplane mode” icon **305**G, and a “Mobile data” icon **305**H, a “Location” icon **305**I, and a “Screenshot” icon **305**J. The quick settings panel **305** may further display one or more other quick settings icons. This is not specifically limited herein.
(51) The “Airplane mode” icon **305**G may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** may disable the airplane mode.
(52) The brightness control **306** may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** may adjust brightness of the display.
(53) For example, as shown in FIG. **4**A and FIG. **4**B, the user taps the “Airplane mode” icon **305**G with a finger, and the user equipment **200** detects the user operation. In response to the user operation, the user equipment **200** displays a no-signal indicator **301**D of a mobile communication signal (which may alternatively be referred to as a cellular signal) on the status bar. It may be learned from the no-signal indicator **301**D in FIG. **4**B that the user equipment **200** has not accessed a network, that is, has not successfully registered with a network.
(54) When the user equipment **200** supports both 4G and 5G networks, and a SA 5G service is not provisioned for the user equipment **200**, the user equipment **200** shown in FIG. **4**A and FIG. **4**B may register with the 4G network after maintaining a mobile signal-free state for several seconds.
(55) For example, as shown in FIG. **5**A and FIG. B, after maintaining a mobile signal-free state for several seconds, the user equipment **200** displays, in the status bar, one or more signal strength indicators **301**E and an operator name (for example, “China Mobile”) **301**F that are of a 4G mobile communication signal, and may further display one or more signal strength indicators **301**E of a wireless fidelity (wireless fidelity, Wi-Fi) signal.
(56) In addition to the manner of disabling the airplane mode shown in FIG. **4**A and FIG. **4**B, in this embodiment of this application, the airplane mode may be disabled in another manner. This is not specifically limited herein.
(57) FIG. **6** shows an example of a user interface **12** that is on the user equipment **200** and that is used to show that the user equipment **200** is restarted and powered on.
(58) The user interface **12** may include: a power-off control **401** and a restart control **402**.
(59) The power-off control **401** may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** may be powered off.
(60) The restart control **402** may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** may be powered off and then powered on again.
(61) For example, as shown in FIG. **7**A and FIG. **7**B, the user taps the restart control **402** with a finger, and the user equipment **200** detects the user operation. In response to the user operation, the user equipment **200** is powered off and then powered on again. After the user equipment **200** is powered on, the user equipment **200** displays a user interface **10**.
(62) Different from the user interface **10** shown in FIG. **4**A and FIG. **4**B, a status bar **201** of the

user interface **10** shown in FIG. **6** may include: a no-signal indicator **201**D, a battery status indicator **201**B, and a time indicator **201**C.
(63) It may be understood that, when the user equipment **200** is just powered on, the user equipment **200** does not access a network. Therefore, the status bar **201** displays the no-signal indicator **201**D. When the user equipment **200** supports both 4G and 5G networks, and a SA 5G service is not provisioned for the user equipment **200**, the user equipment **200** shown in FIG. **6** may register with the 4G network after maintaining a mobile signal-free state for several seconds.
(64) For example, as shown in FIG. **8**A and FIG. **8**B, after maintaining a mobile signal-free state for several seconds, the user equipment **200** displays, in the status bar **201**, one or more signal strength indicators **201**E and an operator name **201**F that are of a 4G mobile communication signal, and may further display one or more signal strength indicators **201**G of a Wi-Fi signal.
(65) It may be understood that when the user equipment **200** supports the 5G network, and the SA 5G service is provisioned before the airplane mode is disabled (or the user equipment **200** is restarted and powered on), the user equipment **200** may register with the SA 5G network after the airplane mode is disabled (or the user equipment **200** is restarted and powered on).
(66) When a user intends to register with the SA 5G network, SA 5G service provisioning needs to be completed in advance via an operator. For a SA 5G network registration request from a user for whom the SA 5G service is not provisioned, different operators may process the request in different ways. Currently, existing processing manners may include the following manners. (1) After a network device receives a SA 5G registration request sent by a mobile phone for which a SA 5G service is not provisioned, the network device sends a registration reject response to the mobile phone, and adds a reject cause value #27 to the message. After receiving the reject cause value #27, the mobile phone cannot initiate a SA 5G network registration request within 12 to 24 hours.
(67) The value #27 indicates that an N1 mode (N1 mode) is not supported, and the value may be used to reject a network registration request from an unprovisioned user. According to a protocol definition, after receiving the rejection cause value #27, the mobile phone disables the SA 5G service for 12 to 24 hours. In other words, the mobile phone cannot initiate a SA 5G network registration request within 12 to 24 hours. (2) After a network device receives a SA 5G registration request sent by a mobile phone for which a SA 5G service is not provisioned, the network device sends a registration reject response to the mobile phone, and adds a reject cause value #111 to the message. After receiving the reject cause value #111, the mobile phone cannot initiate a SA 5G network registration request within a preset time period.
(68) The value #111 indicates a protocol error, and the value may be used to reject a registration request for the first time before automatic provisioning. According to a protocol definition, after receiving the reject cause value #111, the mobile phone cannot register with the SA 5G network within the preset period.
(69) It should be noted that, currently, after rejecting a SA 5G registration request from a user for whom a SA 5G service is not provisioned, some operators may automatically provision the SA 5G service for the user. (3) After a network device receives a SA 5G registration request sent by a mobile phone for which a SA 5G service is not provisioned, the network device sends a registration reject response to the mobile phone, and adds a reject cause value #111 to the message. According to a protocol definition, after receiving the reject cause value #111, the mobile phone triggers a timer T**3052**. Before the timer T**3502** expires, the mobile phone cannot re-initiate a SA 5G network registration request. (4) After a network device receives a SA 5G registration request sent by a mobile phone for which a SA 5G service is not provisioned, the network device sends a registration reject response to the mobile phone, and automatically completes SA 5G service provisioning for the user. After the auto-provisioning is complete, a 4G network side sends a detach request (for example, Detach Request) to the mobile phone, to trigger the mobile phone to re-initiate a 4G network registration request. Therefore, after the auto-provisioning is complete, the mobile phone cannot directly initiate a registration request on the 5G network.

(70) In conclusion, after the network device rejects the SA 5G registration request from a mobile phone that does not register with the SA 5G network, the mobile phone cannot quickly register with the SA 5G network even if the network device automatically completes SA 5G service provisioning for the mobile phone. After the network side completes auto-provisioning for the mobile phone, the mobile phone may need to be powered on again, an airplane mode may need to be disabled, or the network may need to be disconnected, to trigger the mobile phone to search for and register with the SA 5G network again.
(71) To resolve the foregoing problem, an embodiment of this plication provides a network search method for SA 5G service auto-provisioning. In this embodiment of this application, for user equipment **200** for which a SA 5G service is not provisioned, after a network side rejects a SA 5G registration request from the user equipment **200**, an operator completes SA 5G service auto-provisioning for the user equipment **200**. After the auto-provisioning, the user equipment **200** determines, based on specific trigger information, that the SA 5G service provisioning for the user equipment is completed. The user equipment **200** initiates SA 5G network search and registration again, to quickly register with the SA 5G network. According to the method provided in this embodiment of this application, a user for whom a SA 5G service is not provisioned can quickly register with the SA 5G network, so that user experience is improved.
(72) FIG. **9** is a schematic flowchart of a network search method for SA 5G service auto-provisioning according to an embodiment of this application. As shown in FIG. **9**, the network search method for SA 5G service auto-provisioning provided in this embodiment of this application includes but is not limited to step S**601** to step S**607**. The following further describes a possible implementation of the method embodiment. S**601**: User equipment **200** sends a registration request to a first network device, and the first network device receives the registration request sent by the user equipment **200**, where the registration request is used to request to register with a SA 5G network.
(73) In this embodiment of this application, a SA 5G service is not provisioned for the user equipment **200**, and the user equipment **200** supports 4G network and 5G networks. In addition to the 4G and SA 5G networks, the user equipment **200** may further support 3G and 2G networks. When no network service is available, the user equipment **200** preferentially initiates a 5G network search. When the user equipment **200** is in coverage of a 5G cell, the user equipment **200** may find the 5G network, and initiate a SA 5G registration request to a 5G network side.
(74) It may be understood that, in a default setting of the mobile phone, the mobile phone preferably searches for a highest-standard network. For example, when the mobile phone supports 3G, 4G, and 5G networks, the mobile phone first searches for a 5G network. A user may also select, in settings of the mobile phone, a network mode searched first.
(75) In some embodiments of this application, the foregoing registration request may be a SA 5G network registration request sent by the user equipment **200** to the first network device for the first time.
(76) In some implementations of this application, the foregoing registration request may be a SA 5G network registration request sent, when no network service is available in an NR cell, by the user equipment **200** to the first network device for the first time.
(77) For example, when leaving an LTE cell and entering the NR cell, the user equipment **200** initiates the SA 5G network registration request for the first time; or when an airplane mode is disabled in the NR cell, the user equipment **200** initiates the SA 5G network registration request for the first time; or when user equipment **200** is powered on in the NR cell, the user equipment **200** initiates the SA 5G network registration request for the first time.
(78) Before initiating the current registration request, the user equipment **200** may have initiated a SA 5G registration request, or a SA 5G service may have been provisioned for the user equipment **200**. However, when the user equipment **200** sends this registration request, a SA 5G service is not provisioned for the user equipment **200**. It may be understood that before the user equipment **200**

initiates the current registration request, a SA 5G service may be disabled after being provisioned for the user equipment **200**. This is not specifically limited in this embodiment of this application.
(79) In some embodiments of this application, the first network device may include an AMF.
(80) In some embodiments of this application, the foregoing registration request is "registration request".
(81) In some embodiments of this application, the registration request may include a registration type and an identity of the user equipment **200**.
(82) It should be noted that the registration type is initial registration. For example, the registration type is "initial registration". The identity of the user equipment **200** may be a subscription concealed identifier (subscription concealed identifier. SUCI) or a 5G globally unique temporary UE identifier (5G globally unique temporary ue identity, 5G-GUTI). An objective of using the 5G-GUTI in a 5G system is to reduce use of a permanent identifier of the UE displayed in communication, to improve security. The SUCI is a ciphertext obtained through encrypting a subscription permanent identifier (subscription permanent identifier, SUPI) by using a public key. The SUPI of the user equipment is a unique permanent identity of a user in the 5G network. When the user equipment **200** performs initial registration by using the 5G-GUTI of the user equipment **200**, the user equipment **200** indicates, in a network registration request, related information of a globally unique AMF identifier (globally unique amf ID, GUAMI).
(83) In some implementations of this application, that the user equipment **200** sends the registration request to the first network device includes: the user equipment **200** sends the registration request to an (R)AN, and the (R)AN sends the registration request to the first network device.
(84) In some embodiments of this application, the foregoing registration request may further include an N2 parameter, and the N2 parameter may include a selected public land mobile network (Public Land Mobile Network, PLMN) identity document (Identity document, ID), location information, and a cell identity.
(85) It should be noted that if the registration request includes the 5G-GUTI, the (R)AN may directly determine an AMF based on the GUAMI in the registration request. If the (R)AN cannot select an appropriate AMF, the (R)AN forwards the registration request to an AMF preconfigured by the (R)AN, to perform AMF selection. S**602**: The first network device sends a registration reject response to the user equipment **200**, and the user equipment **200** receives the registration reject response sent by the first network device, where the registration reject response is used to notify the user equipment **200** that the registration with the SA 5G network fails.
(86) In some embodiments of this application, the foregoing registration request is "registration reject".
(87) In some embodiments of this application, the registration reject response includes reject cause information.
(88) In some embodiments of this application, the reject cause information is a reject cause value. For example, the reject cause value is #111 or #27.
(89) In some embodiments of this application, that the first network device sends the registration reject response to the user equipment **200** includes: the first network device sends the registration reject response to the (R)AN, and the (R)AN sends the registration reject response to the user equipment **200**.
(90) In some embodiments of this application, the first network device includes an AMF and a UDM. For example, as shown in FIG. **10**A, after step S**601** and before step S**602**, the method may further include step S**602***a* and step S**602***b*. S**602***a*: The AMF sends a query request to the UDM, and the UDM receives the query request sent by the AMF, Where the query request is used to request to query SA 5G subscription data of the user equipment **200**.
(91) In some embodiments of this application, the query request includes the identity of the user equipment **200**. For example, the query request includes an international mobile subscriber identity (International Mobile SubscriberIdentification Number, IMSI) or the SUCI. The IMSI is an identity

used to distinguish different users in a cellular network, and the identity is unique in all cellular networks.
(92) In some embodiments of this application, the UDM stores SA 5G subscription data of one or more pieces of user equipment. A record of SA 5G subscription data of each user equipment includes an identity of the user equipment. The UDM may query, based on an identity of user equipment, whether the UDM stores SA 5G subscription data of the user equipment.
(93) In some embodiments of this application, after step S**601**, the AMF invokes an AUSF to attempt to authenticate the identity of the user equipment **200**. When the authentication succeeds, the AMF performs step S**602***a*.
(94) In some embodiments of this application, the foregoing query request may be "Nudm_UECM_Registration". S**602***b*: The UDM sends a query failure response to the AMF, and the AMF receives the query failure response sent by the UDM, where the query failure response indicates that the user equipment **200** has no available SA 5G subscription data.
(95) In some embodiments of this application, after the user equipment **200** fails to register with the SA 5G network, the user equipment **200** searches for the 4G network and initiates a 4G network registration request. Therefore, for example, as shown in FIG. **10**A, after step S**602**, the method may further include step S**602***c* and step S**602***d*. S**602***c*: The user equipment **200** sends the 4G registration request to a mobility management entity (mobility management entity, MME), and the MME receives the 4G registration request sent by the user equipment **200**, where the 4G registration request is used to request to register with the 4G network.
(96) Specifically, the user equipment **200** sends the 4G registration request to the MME through a 4G access network. The 4G access network may be an eNB.
(97) The MME is a key control node in the 4G network, and is mainly responsible for functions such as mobility management, bearer management, user authentication, and selection of a serving gateway and a public data network (public data network, PDN) gateway. S**602***d*: The MME sends a 4G registration accept response to the user equipment **200**, and the user equipment **200** receives the 4G registration accept response sent by the MME, where the 4G registration accept response is used to notify the user equipment **200** that the registration with the 4G network succeeds.
(98) In some embodiments of this application, the MME sends the 4G registration request to the user equipment **200** through the 4G access network, S**603**: The first network device sends a provisioning request to a maintenance system of an operator, and the maintenance system of the operator receives the provisioning request sent by the first network device, where the provisioning request is used to request to provision the SA 5G service for the user equipment **200**.
(99) In some embodiments of this application, the provisioning request includes the identity of the user equipment **200**. For example, the provisioning request includes the IMSI.
(100) In some embodiments of this application, after receiving the provisioning request sent by the first network device, the maintenance system of the operator sends a reception acknowledgment message to the first network device, where the reception acknowledgment message is used to notify the first network device that the maintenance system has received the provisioning request.
(101) In some embodiments of this application, after the maintenance system receives the provisioning request sent by the first network device, when the maintenance system allows the SA 5G service to be automatically provisioned for the user equipment **200**, the maintenance system notifies a home subscriber server (home subscriber server, HSS) of sending subscription data of the user equipment **200** stored in the HSS to the first network device. After obtaining the subscription data of the user equipment **200**, the first network device sends an obtaining notification message to the maintenance system of the operator, to notify the maintenance system that the first network device has obtained the subscription data of the user equipment **200**.
(102) The HSS is a server configured to store user subscription data in a 4G evolved packet system (evolved packet system, EPS), and is mainly responsible for managing the user subscription data and location information of a mobile user. The subscription data stored in the HSS mainly includes:

user identity information (identity, number, and address), user security context information (authentication and authorization information for user network access control), user location information, and user service subscription information (including value-added service data of another application server). It may be understood that the HSS may store a parameter used to identify whether a user subscribes to a 4G service.
(103) It should be noted that after obtaining the subscription data of the user equipment **200**, the first network device may modify or delete information in the subscription data of the user equipment **200** based on a user requirement, to generate SA 5G subscription data of the user equipment **200**. After the SA 5G subscription data of the user equipment **200** is generated, the first network device may store a parameter used to identify whether the user equipment **200** subscribes to a SA 5G service. It may be understood that when the first network device stores the SA 5G subscription data of the user equipment **200**, and the foregoing parameter represents that the user equipment **200** subscribes to the SA 5G service, the SA 5G subscription data is available SA 5G subscription data of the user equipment **200**.
(104) In some embodiments of this application, the first network device includes an AMF and a UDM, and the maintenance system notifies the HSS of sending the subscription data of the user equipment **200** stored in the HSS to the UDM. The UDM may generate the SA 5G subscription data of the user equipment **200** based on the subscription data of the user equipment **200** sent by the HSS.
(105) In some embodiments of this application, after sending the subscription data of the user equipment **200** to the first network device, the HSS may delete the subscription data of the user equipment **200** from the HSS, or may not delete the subscription data of the user equipment **200** from the HSS. This is not specifically limited herein.
(106) In some embodiments of this application, the first network device includes an AMF and a UDM. After (or before, or when) the UDM sends a query failure response to the AMF, the UDM may send a provisioning request to the maintenance system of the operator. For example, as shown in FIG. **10**A, after step S**602**, the method may further include step S**603***a*. S**603***a*: The UDM network device sends the provisioning request to the maintenance system of the operator, and the maintenance system of the operator receives the provisioning request sent by the UDM, where the provisioning request is used to request to provision the SA 5G service for the user equipment **200**. S**604**: After the maintenance system of the operator determines that the SA 5G service is successfully provisioned for the user equipment **200**, the maintenance system of the operator sends a notification short message to the user equipment **200**, and the user equipment **200** receives the notification short message sent by the maintenance system of the operator, where the notification short message is used to notify the user equipment **200** that the SA 5G service is successfully provisioned.
(107) It should be noted that, after the maintenance system of the operator receives an obtaining notification message sent by the first network device, and determines, through the obtaining notification message, that the SA 5G service is successfully provisioned for the user equipment **200**, the maintenance system of the operator may send the notification short message to the user equipment **200** by using one or more network devices.
(108) In some embodiments of this application, after the first network device receives the subscription data of the user equipment **200** sent by the HSS, and before step S**604**, the method further includes: the first network device sends acknowledgment information to the maintenance system of the operator; the maintenance system of the operator receives the acknowledgment information sent by the first network device, where the acknowledgment information is used to indicate that the first network device has obtained the SA 5G subscription data of the user equipment **200**; and the maintenance system of the operator determines, based on the acknowledgment information, that the SA 5G service is successfully provisioned for the user equipment **200**.

(109) For example, the user equipment **200** receives a SA 5G service provisioning notification short message sent by the operator. As shown in FIG. **11**, a user interface **13** is configured to display the SA 5G service provisioning notification short message received by the user equipment **200**.
(110) The user interface **13** may include: a status bar **901**, a navigation bar **902**, a return control **903**, a sender number **904**, an editing control **905**, a short message receiving time point **906**, a profile picture **907**, a short message content display area **908**, a voice control **909**, an input box **910**, and a sending control **911**.
(111) The return control **903** may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** may display a user interface previous to the user interface **13**.
(112) The editing control **905** may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** may receive an editing operation performed by a user on a display card **1608**, for example, deletion and forwarding.
(113) The voice control **909** may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** begins to collect a voice.
(114) The input box **910** may receive a user operation (for example, a touch operation). In response to the detected user operation, the user equipment **200** may display an input keyboard. The input box **910** may be configured to display input content.
(115) The short message content display area **908** is configured to display specific content of a notification short message. For example, the specific content of the notification short message may include the following: "Dear user, a SA 5G service has been successfully provisioned for you. If you have any question, call the customer service hotline **10008**. Thank you for using China Mobile. Have a nice day."
(116) In some embodiments of this application, before the maintenance system of the operator sends the notification short message to the user equipment **200**, the method further includes: the maintenance system of the operator sends a confirmation request short message to the user equipment **200**; the user equipment **200** receives the confirmation request short message sent by the maintenance system of the operator, where the confirmation request short message is used to request the user to confirm whether the SA 5G service is to be provisioned; and the user equipment **200** sends a confirmation response short message to the maintenance system of the operator, where the confirmation response short message is used to indicate that the user equipment **200** confirms that the SA 5G service is to be provisioned.
(117) For example, as shown in FIG. **12** the user equipment **200** receives a confirmation request short message sent by an operator. The user interface **13** may further include: a short message receiving time point **912**, a profile picture **913**, and a short message content display area **914**.
(118) The short message content display area **913** is configured to display specific content of the confirmation request short message. For example, the specific content of the confirmation request short message may include the following: "Dear user, thank you for using China Mobile. In return for your support, a SA 5G service can be provisioned for you. Reply "Y" to confirm the provisioning. If the provisioning is not required, ignore the message. Have a nice day."
(119) For example, as shown in FIG. **13**A and FIG. **13**B, the user equipment **200** returns a confirmation response short message based on the confirmation request short message. The user interface **13** may further include: a short message receiving time point **915** and a short message content display area **916**.
(120) The short message content display area **913** is configured to display specific content of the confirmation response short message. For example, it can be learned from the foregoing confirmation request short message that when the user agrees that the SA 5G service is to be provisioned, the specific content of the confirmation response short message may be "Y".
(121) After the operator receives the confirmation response short message returned by the user

equipment **200**, when determining that the SA 5G service is successfully provisioned for the user equipment **200**, the operator sends a SA 5G service provisioning notification short message to the user equipment **200**. As shown in FIG. **14**, after returning the confirmation response short message, the user equipment **200** receives the SA 5G service provisioning notification short message sent by the operator. S**605**: The user equipment **200** parses the notification short message, and determines, based on the notification short message, that the SA 5G service is successfully provisioned for the user equipment **200**.
(122) In some embodiments of this application, the user equipment **200** parses the notification short message by using a neural network.
(123) In some embodiments of this application, after the user equipment **200** is powered on, the user equipment **200** parses whether a received short message is the short message indicating that the SA 5G service is successfully provisioned. When the received short message is the short message indicating that the SA 5G service is successfully provisioned, step S**610** is performed.
(124) In some embodiments of this application, after sending a SA 5G registration request, the user equipment **200** parses whether a received short message is the short message indicating that the SA 5G service is successfully provisioned. When the received short message is the short message indicating that the SA 5G service is successfully provisioned, step S**610** is performed.
(125) In some embodiments of this application, after receiving a SA 5G registration reject response, the user equipment **200** parses whether a received short message is the short message indicating that the SA 5G service is successfully provisioned. When the received short message is the short message indicating that the SA 5G service is successfully provisioned, step S**611** is performed.
(126) In some embodiments, after receiving the short message shown in FIG. **11**, the user equipment **200** may identify the short message by using a trained neural network model, to determine that the short message is the notification short message indicating that the SA 5G service is provisioned.
(127) It may be understood that the user equipment **200** may parse each received short message (or parse each received short message whose sender is the operator) after the user equipment **200** is powered on, sends the SA 5G registration request, or receives the SA 5G registration reject response. When content of the parsed short message includes a related keyword indicating that the SA 5G service is successfully provisioned, it may be determined that the SA 5G service is successfully provisioned for the user equipment **200**. In this way, the user equipment **200** is triggered to initiate SA 5G network search and registration again. S**606**: The user equipment **200** sends a registration request to the first network device again, and the first network device receives the registration request sent by the user equipment **200**, where the registration request is sent by the user equipment **200** after the user equipment **200** determines, based on the notification short message, that the SA 5G service is successfully provisioned for the user equipment **200**. S**607**: The first network device sends a registration accept response to the user equipment **200**, and the user equipment **200** receives the registration accept response sent by the first network device, where the registration accept response is used to notify the user equipment **200** that the registration with the SA 5G network succeeds.
(128) In some implementations of this application, that the first network device sends the registration accept response to the user equipment **200** includes: the first network device sends the registration accept response to the (R)AN, and the (RAN sends the registration accept response to the user equipment **200**.
(129) In some embodiments of this application, the foregoing first registration request is “registration receipt”.
(130) In some embodiments of this application, the registration accept response may include: the 5G-GUTI, a registration area, and allowed network slice selection assistance information (network slice selection assistance information, NSSAI).
(131) In some embodiments of this application, the first network device includes an AMF and a

UDM. For example, as shown in FIG. **10**B, after step S**606** and before step S**607**, the method may further include step S**607***a* and step S**607***b*. S**607***a*: The AMF sends a query request to the UDM again, and the UDM receives the query request sent by the AMF, where the query request is used to request to query SA 5G subscription data of the user equipment **200**. S**607***b*: The UDM sends a query success response to the AMF, and the AMF receives the query success response sent by the UDM, where the query success response is used to indicate that the user equipment **200** has available SA 5G subscription data.
(132) It may be understood that the AMF sends the registration accept response to the user equipment **200** only after the AMF receives the query success response sent by the UDM and determines that the user equipment **200** has the available SA 5G subscription data. The query success response may include the SA 5G subscription data of the user equipment **200**, or may not include the SA 5G subscription data of the user equipment **200**. This is not specifically limited herein. In this embodiment of this application, the available SA 5G subscription data may alternatively be referred to as available 5G subscription data for short.
(133) In this embodiment of this application, a sequence of step S**602** and step S**603** is not specifically limited. It may be understood that step S**602** may be performed before or after step S**603**, or may be performed simultaneously with step S**603**.
(134) FIG. **15**A and FIG. **15**B are a schematic flowchart of still another network search method for SA 5G service auto-provisioning according to an embodiment of this application. As shown in FIG. **15**A and FIG. **15**B, the network search method for SA 5G service auto-provisioning provided in this embodiment of this application includes but is not limited to step S**701** to step S**707**. The following further describes a possible implementation of the method embodiment.
(135) S**701**: User equipment **200** sends a registration request to a first network device, and the first network device receives the registration request sent by the user equipment **200**, where the registration request is used to request to register with a SA 5G network.
(136) In some embodiments of this application, the first network device includes an AMF.
(137) In some embodiments of this application, the foregoing registration request is “registration request”.
(138) In some implementations of this application, that the user equipment **200** sends the registration request to the first network device includes: the user equipment **200** sends the registration request to an (JOAN, and the (R)AN sends the registration request to the first network device.
(139) S**702**: The first network device sends a registration reject response to the user equipment **200**, and the user equipment **200** receives the registration reject response sent by the first network device, where the registration reject response includes reject cause information, and the registration reject response is used to notify the user equipment **200** that registration with the SA 5G network fails, and the reject cause information is used to indicate a cause of the registration failure.
(140) In some embodiments of this application, that the first network device sends the registration reject response to the user equipment **200** includes: the first network device sends the registration reject response to the (R)AN, and the (R)AN sends the registration reject response to the user equipment **200**.
(141) In some embodiments of this application, the foregoing registration request is “registration reject”.
(142) In some embodiments of this application, the registration reject response includes reject cause information.
(143) In some embodiments of this application, the reject cause information is a reject cause value. For example, the reject cause value is #111.
(144) In some embodiments of this application, the first network device includes an AMF and a UDM. For example, as shown in FIG. **16**A, after step S**701** and before step S**702**, the method may further include step S**702***a* and step S**702***b*. S**702***a*: The AMF sends a query request to the UDM,

and the UDM receives the query request sent by the AMF, where the query request is used to request to query SA 5G subscription data of the user equipment **200**. S**702***b*: The UDM sends a query failure response to the AMF, and the AMF receives the query failure response sent by the UDM, where the query failure response indicates that the user equipment **200** has no available SA 5G subscription data.

(145) In some embodiments of this application, after the user equipment **200** fails to register with the SA 5G network, the user equipment **200** searches for the 4G network and initiates 4G network registration. Therefore, for example, as shown in FIG. **16**A and FIG. **16**B, after step S**702**, the method may further include step S**702***c* and step S**702***d*. S**702***c*: User equipment **200** sends a 4G registration request to an MME, and the MME receives the 4G registration request sent by the user equipment **200**, where the 4G registration request is used to request to register with the 4G network. S**702***d*: The MME sends a 4G registration accept response to the user equipment **200**, and the user equipment **200** receives the 4G registration accept response sent by the MME, where the 4G registration accept response is used to notify the user equipment **200** that the registration with the 4G network succeeds. S**703**: The first network device sends a provisioning request to a maintenance system of an operator, and the maintenance system of the operator receives the provisioning request sent by the first network device, where the provisioning request is used to request to provision the SA 5G service for the user equipment **200**.

(146) In some embodiments of this application, the provisioning request includes the identity of the user equipment **200**. For example, the provisioning request includes the IMSI.

(147) In some embodiments of this application, the first network device includes an AMF and a UDM. After or before, or when) the UDM sends a query failure response to the AMF, the UDM may send a provisioning request to the maintenance system of the operator. For example, as shown in FIG. **16**A, after step S**702**, the method may further include step S**703***a*. S**703***a*: The UDM network device sends the provisioning request to the maintenance system of the operator, and the maintenance system of the operator receives the provisioning request sent by the UDM where the provisioning request is used to request to provision the SA 5G service for the user equipment **200**. S**704**: The maintenance system of the operator sends a transfer request to an HSS, and the HSS receives the transfer request sent by the maintenance system of the operator, where the transfer request is used to request the HSS to send the subscription data of the user equipment **200** to the first network device.

(148) In some embodiments of this application, the provisioning request includes the identity of the user equipment **200**. For example, the provisioning request includes the IMSI.

(149) It may be understood that after the maintenance system receives the provisioning request, when the maintenance system allows the SA 5G service to be automatically provisioned for the user equipment **200**, the maintenance system notifies the HSS to send the subscription data of the user equipment **200** stored in the HSS to a network device that is of a 5G core network and that is configured to store subscription data. S**705**: After the HSS sends the subscription data of the user equipment **200** to the first network device, the HSS sends a positioning cancel request to the MME, where the positioning cancel request is used to trigger the MME to send a detach request to the UE.

(150) Optionally, the positioning cancel request is further used to request the MME to delete a mobility management (mobility management, MM) context and an EPS bearer that are subscribed to by the user equipment **200**.

(151) It should be noted that after obtaining the subscription data of the user equipment **200**, the first network device may modify or delete information in the subscription data of the user equipment **200** based on a user requirement, and determine SA 5G subscription data of the user equipment **200**.

(152) In some embodiments of this application, after sending the subscription data of the user equipment **200** to the first network device, the HSS may delete the subscription data of the user equipment **200** from the HSS, or may not delete the subscription data of the user equipment **200**

from the HSS. This is not specifically limited herein.
(153) In some embodiments of this application, the foregoing positioning cancel request is “Cancel Location”.
(154) In some embodiments of this application, the positioning cancel request includes a cancellation type (cancellation type) and an identity of the user equipment **200** (for example, an IMSI). The cancellation type is “subscription withdrawn”.
(155) The MM context is a set of information stored in an MS and an SGSN, and includes a user identity, status information, location information, and a terminal capability. A user can have only one MM context.
(156) The EPS bearer provides a specific QoS guarantee for transmission between a UE and a PDN, and there are two types of EPS bearers: a default bearer and a dedicated bearer. One EPS bearer is a logical aggregation of one or more service data flows (service data flow, SDF) between the UF, and the PDN.
(157) In some embodiments of this application, the first network device includes an AMF and a UDM. For example, as shown in FIG. **16**B, before step S**705**, the proposed solution further includes S**705***a*. S**705***a*: The HSS sends the subscription data of the user equipment **200** to the UDM, and the UDM receives the subscription data of the user equipment **200** sent by the HSS. S**706**: The MME sends a detach request to the user equipment **200**, where the detach request is used to trigger the user equipment **200** to re-register with the 4G network.
(158) In some embodiments of this application, the foregoing detach request is “Detach Request”.
(159) In some embodiments of this application, the detach request carries the identity of the user equipment **200**. For example, the detach request carries the IMSI.
(160) In some embodiments of this application, the detach request may further include a detach type and a detach cause.
(161) In some embodiments of this application, the detach request is sent by the MME after the MME deletes the MM context and the EPS bearer that are subscribed to by the user equipment **200**.
(162) In some embodiments of this application, for example, as shown in FIG. **16**C, before step S**706**, the proposed solution further includes S**706***a*. S**706***a*: The MME deletes the MM context and the EPS bearer that are subscribed to by the user equipment **200**. S**707**: The user equipment **200** sends a registration request to the first network device again, and the first network device receives the registration request sent by the user equipment **200**, where the registration request is sent when the user equipment **200** receives the detach request after receiving the cause information. S**708**: The first network device sends a registration accept response to the user equipment **200**, and the user equipment **200** receives the registration accept response sent by the first network device, where the registration accept response is used to notify the user equipment **200** that the registration with the SA 5G network succeeds.
(163) In some implementations of this application, that the first network device sends the registration accept response to the user equipment **200** includes: the first network device sends the registration accept response to the (R)AN, and the (R)AN sends the registration accept response to the user equipment **200**.
(164) In some embodiments of this application, the first network device includes an AMF and a UDM. For example, as shown in FIG. **16**C, after step S**708** and before step S**709**, the method may further include step S**709***a* and step S**709***b*. S**708***a*: The AMF sends a query request to the UDM again, and the UDM receives the query request sent by the AMF, where the query request is used to request to query SA 5G subscription data of the user equipment **200**. S**708***b*: The UDM sends a query success response to the AMF, and the AMF receives the query success response sent by the UDM, where the query success response is used to indicate that the user equipment **200** has available SA 5G subscription data.
(165) It may be understood that after receiving the subscription data of the user equipment **200** sent by the HSS, the UDM may determine the SA 5G subscription data of the user equipment **200** based

on the subscription data of the user equipment **200**. Therefore, after the UDM receives the query request sent by the AMF in step S**708***a*, the UDM may obtain the available SA 5G subscription data of the user equipment **200** through query.
(166) In this embodiment of this application, a sequence of step S**702** and step S**703** is not specifically limited.
(167) In this embodiment of this application, for specific implementations of step S**701** to step S**703**, refer to the optional embodiments of step S**601** to step S**603**. For specific implementations of step S**707** to step S**708**, refer to step S**606** and the optional embodiments of step S**606**. Details are not described herein again.
(168) FIG. **17** is a schematic flowchart of a network search method for SA 5G service auto-provisioning according to an embodiment of this application. As shown in FIG. **17**, the network search method for SA 5G service auto-provisioning provided in this embodiment of this application includes but is not limited to step S**801** to step S**805**. The following further describes a possible implementation of the method embodiment. S**801**: User equipment **200** sends a registration request to a first network device, and the first network device receives the registration request sent by the user equipment **200**, where the registration request is used to request to register with a SA Ski network.
(169) In some embodiments of this application, the first network device includes an AMF.
(170) In some embodiments of this application, the foregoing registration request is “registration request”.
(171) In some implementations of this application, that the user equipment **200** sends the registration request to the first network device includes: the user equipment **200** sends the registration request to an (R)AN, and the (R)AN sends the registration request to the first network device. S**802**: The first network device sends a registration reject response to the user equipment **200**, and the user equipment **200** receives the registration reject response sent by the first network device, where the registration reject response includes reject cause information, and the registration reject response is used to notify the user equipment **200** that registration with the SA 5G network fails, and the reject cause information is used to indicate a cause of the registration failure.
(172) In some embodiments of this application, that the first network device sends the registration reject response to the user equipment **200** includes: the first network device sends the registration reject response to the (R)AN, and the (R)AN sends the registration reject response to the user equipment **200**.
(173) In some embodiments of this application, the foregoing registration request is “registration reject”.
(174) In some embodiments of this application, the registration reject response includes reject cause information.
(175) In some embodiments of this application, the reject cause information is a reject cause value. For example, the reject cause value is #111.
(176) In some embodiments of this application, the first network device includes an AMF and a UDM. For example, as shown in FIG. **18**A, after step S**801** and before step S**802**, the method may further include step S**802***a* and step S**802***b*. S**802***a*: The AMF sends a query request to the UDM, and the UDM receives the query request sent by the AMF, where the query request is used to request to query SA 5G subscription data of the user equipment **200**. S**802***b*: The UDM sends a query failure response to the AMF and the AMF receives the query failure response sent by the UDM, where the query failure response indicates that the user equipment **200** has no available SA 5G subscription data.
(177) In some embodiments of this application, after the user equipment **200** fails to register with the SA 5G network, the user equipment **200** searches for the 4G network and initiates 4G network registration. Therefore, for example, as shown in FIG. **18**A and FIG. **18**B, after step S**802**, the method may further include step S**802***c* and step S**802***d*. S**802***c*: User equipment **200** sends a 4G

registration request to an MME, and the MME receives the 4G registration request sent by the user equipment **200**, where the 4G registration request is used to request to register with the 4G network. S**802***d*: The MME sends a 4G registration accept response to the user equipment **200**, and the user equipment **200** receives the 4G registration accept response sent by the MME, where the 4G registration accept response is used to notify the user equipment **200** that the registration with the 4G network succeeds.

(178) It may be understood that after receiving, the SA 5G registration reject response, the user equipment **200** disables a 5G network search capability, searches for the 4G network, and initiates 4G network registration. The 4G access network may be an eNB. S**803**: After receiving the reject cause information, the user equipment **200** starts a timer T**3502**, where timing duration of the T**3502** is a preset timing value.

(179) It should be noted that default timing duration of T**3502** is 12 minutes, but actual SA 5G service provisioning duration is usually much shorter than 12 minutes.

(180) In some embodiments of this application, actual SA 5G service provisioning duration is measured for a plurality of times, to calculate average SA 5G service provisioning duration. The preset timing value is the average provisioning duration.

(181) In some embodiments of this application, SA 5G service provisioning duration in different networks is measured, to calculate average SA 5G service provisioning duration in the different networks. The preset timing value is, based on each different network, the average SA 5G service provisioning duration corresponding to the network. S**804**: The first network device sends a provisioning request to a maintenance system of an operator, and the maintenance system of the operator receives the provisioning request sent by the first network device, where the provisioning request is used to request to provision the SA 5G service for the user equipment **200**.

(182) In some embodiments of this application, the provisioning request includes the identity of the user equipment **200**. For example, the provisioning request includes an IMSI or a SUCI.

(183) In some embodiments of this application, the first network device includes an AMF and a UDM. After (or before, or when) the UDM sends the query failure response to the first network device, the UDM may send the provisioning request to the maintenance system of the operator. For example, as shown in FIG. **18**A, the solution provided in this embodiment of this application may include S**804***a*. S**804***a*: The UDM network device sends the provisioning request to the maintenance system of the operator, and the maintenance system of the operator receives the provisioning request sent by the UDM, where the provisioning request is used to request to provision the SA 5G service for the user equipment **200**.

(184) In some embodiments of this application, after the UDM sends the query failure response to the AMF, the AMF or the UDM sends the provisioning request to the maintenance system of the operator. After the maintenance system receives the provisioning request, when the maintenance system allows the SA 5G service to be automatically provisioned for the user equipment **200**, the maintenance system notifies the HSS to send the subscription data of the user equipment **200** stored in the HSS to the first network device. After obtaining the subscription data of the user equipment **200**, the first network device may determine the SA 5G subscription data of the user equipment **200** based on the subscription data of the user equipment **200**. Then, the first network device may send an obtaining notification message to the maintenance system of the operator, to notify the maintenance system that the first network device has obtained the SA 5G subscription data of the user equipment **200**. S**805**: The user equipment **200** sends a registration request to the first network device again, and the first network device receives the registration request sent by the user equipment **200**, where the registration request is sent by the user equipment **200** after the T**3502** expires. S**806**: The first network device sends a registration accept response to the user equipment **200**, and the user equipment **200** receives the registration accept response sent by the first network device, where the registration accept response is used to notify the user equipment **200** that the registration with the SA 5G network succeeds.

(185) In some implementations of this application, that the first network device sends the registration accept response to the user equipment **200** includes: the first network device sends the registration accept response to the (R)AN, and the (R)AN sends the registration accept response to the user equipment **200**.
(186) In some embodiments of this application, the first network device includes an AMF and a UDM. For example, as shown in FIG. **18**B, after step S**708** and before step S**709**, the method may further include step S**709***a* and step S**709***b*. S**806***a*: The AMF sends a query request to the UDM again, and the UDM receives the query request sent by the AMF, where the query request is used to request to query SA 5G subscription data of the user equipment **200**. S**806***b*: The UDM sends a query success response to the AMF, and the AMF receives the query success response sent by the UM where the query success response is used to indicate that the user equipment **200** has available SA 5G subscription data.
(187) In this embodiment of this application, a sequence of step S**802** and step S**803** is not specifically limited.
(188) In this embodiment of this application, for specific implementations of step S**801** to step S**803**, refer to the optional embodiments of step S**601** to step S**603**. For specific implementations of step S**805** to step S**806**, refer to step S**606** and the optional embodiments of step S**606**. Details are not described herein again.
(189) It should be noted that in the network search methods for SA 5G service auto-provisioning that are shown in FIG. **9** and FIG. **10**A and FIG. **10**B, one or more of step S**704** to step S**706** may alternatively be included after step S**603** or S**603***a*. This is not specifically limited herein.
(190) In some embodiments of this application, for the network search methods shown in FIG. **9** and FIG. **10**A and FIG. **10**B, one or more of step S**704** to step S**706** are included after step S**603** or step S**603***a*. When the user equipment **200** receives, before step S**605**, the detach request sent by the MME (step S**706**), the user equipment **200** re-initiates 4G registration, and registers with the 4G network. Then, in step S**605**, when the user equipment **200** determines, based on the notification short message, that the SA 5G service provisioning is completed, the user equipment **200** disconnects from the 4G network, and performs step S**606**, that is, re-initiates SA 5G network registration. Before step S**706**, when the user equipment **200** determines, based on the notification short message, that the SA 5G service provisioning is completed (step S**705**), the user equipment **200** disconnects from the current 4G network, and re-initiates SA 5G network registration. It may be understood that in this case, the user equipment **200** may register with the SA 5G network, and no longer perform step S**706**.
(191) It should be noted that in the network search methods for SA 5G service auto-provisioning that are shown in FIG. **17** and FIG. **18**A and FIG. **18**B, one or more of step S**704** to step S**706** may alternatively be included after step S**803** or step S**803***a*. This is not specifically limited herein.
(192) In some embodiments of this application, for the network search methods shown in FIG. **17** and FIG. **18**A and FIG. **18**B, one or more of step S**704** to step S**706** are included after step S**803** or step S**803***a*. When the user equipment **200** receives, before the T**3502** expires, the detach request sent by the MME (step S**706**), the user equipment **200** re-initiates 4G registration, and registers with the 4G network. Then, when the T**3502** expires, the user equipment **200** disconnects from the 4G network, and performs step S**805**, that is, re-initiates SA 5G network registration. When the T**3502** expires before step S**706** is performed, the user equipment **200** disconnects from the current 4G network, and re-initiates SA 5G network registration. It may be understood that in this case, the user equipment **200** may register with the SA 5G network, and no longer perform step S**706**.
(193) It should be noted that in the network search methods for SA 5G service auto-provisioning that are shown in FIG. **16**A, FIG. **16**B, and FIG. **16**C to FIG. **18**A and FIG. **18**B, after the maintenance system of the operator automatically provisions the SASE service for the user equipment **200**, the maintenance system of the operator may also send a notification short message to the user equipment **200** through the (R)AN, to notify the user equipment **200** that the SA 5G

service provisioning succeeds. This is not specifically limited herein.
(194) It can be learned from the network search methods for SA 5G service auto-provisioning shown in FIG. **9**, FIG. **10**A and FIG. **10**B, and FIG. **15**A and FIG. **15**B to FIG. **18**A and FIG. **18**B that, in the network search methods provided in embodiments of this application, after an operator successfully performs auto-provisioning for the user equipment **200**, specific trigger information (for example, existing information in an existing protocol) is used to notify the user equipment **200** that SA 5G service provisioning is completed, to trigger the user equipment **200** to initiate SA 5G network search and SA 5G network registration again. The first trigger information in the network search methods shown in FIG. **9** and FIG. **10**A and FIG. **10**B is to determine, by parsing a short message, that SA 5G service provisioning is completed. The second trigger information in the network search methods shown in FIG. **15**A and FIG. **15**B and FIG. **16**A, FIG. **16**B, and FIG. **16**C is that the reject cause information and the detach request are received. The third trigger information in the network search methods shown in FIG. **17** and FIG. **18**A and FIG. **18**B is that the T**3502** expires after the reject cause information is received.
(195) In some embodiments of this application, at least two of the network search methods for SA 5G service auto-provisioning that are shown in FIG. **9** (or FIG. **10**A and FIG. **10**B), FIG. **15**A and FIG. **15**B (or FIG. **16**A, FIG. **16**B, and FIG. **16**C), and FIG. **17** (or FIG. **18**A and FIG. **18**B) may be alternatively used, to enable a user for whom a SA 5G service is not provisioned to quickly register with a SA 5G network.
(196) For example, the network search methods for SA 5G service auto-provisioning that are shown in FIG. **9** and FIG. **15**A and FIG. **15**B are used.
(197) In some embodiments of this application, for the network search method shown in FIG. **9** (or FIG. **10**A and FIG. **10**B), one or more of step S**704** to step S**706** may be included after step S**603**, in this case, after the user equipment **200** receives the registration reject response in step S**602**, when the user equipment **200** determines at least one of the first trigger information and the second trigger information, the user equipment **200** performs step S**606**, that is, re-initiates SA 5G network registration.
(198) For example, the network search methods for SA 5G service auto-provisioning that are shown in FIG. **9** and FIG. **17** are used.
(199) In some embodiments of this application, for the network search method shown in FIG. **9** (or FIG. **10**A and FIG. **10**B), the registration reject response message may also carry reject cause information, and step S**804** may be included after step S**603**. In this case, after the user equipment **200** receives the registration reject response in step S**602**, when the user equipment **200** determines at least one of the second trigger information and the third trigger information, the user equipment **200** performs step S**606**, that is, re-initiates SA 5G network registration.
(200) For example, the network search methods for SA 5G service auto-provisioning that are shown in FIG. **15**A and FIG. **15**B and FIG. **17** are used.
(201) In some embodiments of this application, for the network search method shown in FIG. **15**A and FIG. **15**B (or FIG. **16**A, FIG. **16**B, and FIG. **16**C), the registration reject response message may also carry reject cause information, and step S**804** may be included after step S**703**. In this case, after the user equipment **200** receives the registration reject response in step S**702**, when the user equipment **200** determines at least one of the second trigger information and the third trigger information, the user equipment **200** performs step S**707**, that is, re-initiates SA 5G network registration.
(202) For example, the network search methods for SA 5G service auto-provisioning that are shown in FIG. **9**, FIG. **15**A and FIG. **15**B, and FIG. **17** are used.
(203) In some embodiments of this application, for the network search method shown in FIG. **9** (or FIG. **10**A and FIG. **10**B), the registration reject response message may also carry reject cause information. One or more of step S**704** to step S**706** may be included after step S**603**, and step S**804** may be included. In this case, after the user equipment **200** receives the registration reject

response in step S**602**, when the user equipment **200** determines at least one of the first trigger information, the second trigger information, and the third trigger information, the user equipment **200** performs step S**606**, that is, re-initiates SA 5G network registration.
(204) In conclusion, in this embodiment of this application, for user equipment for which a SA 5G service is not provisioned, after a network device rejects a SA 5G registration request from the user equipment, an operator completes SA 5G service auto-provisioning for the user equipment. After determining, based on the specific trigger information, that the SA 5G service auto-provisioning has been completed, the user equipment initiates SA 5G network search and SA 5G network registration again. Therefore, according to the solution provided in this embodiment of this application, the user equipment can quickly register with the SA 5G network without being powered on again, disabling an airplane mode, or being disconnected from a network, so that user experience is effectively improved. In addition, in the solution provided iii this embodiment of this application, an existing protocol does not need to be modified, and modification is made only on a user equipment side. Therefore, the solution provided in this embodiment of this application has high applicability.
(205) It should be noted that, in this application, the network side device may include the foregoing first network device and the maintenance system of the operator, the SA 5G network may alternatively be referred to as a 5G network for short, the SA 5G subscription data may alternatively be referred to as 5G subscription data for short, and the SA 5G service may alternatively be referred to as a 5G service for short.
(206) For ease of understanding beneficial effects of the solution provided in this embodiment of this application, the following describes some example graphical user interfaces implemented on the user equipment **200** provided in this embodiment of this application by using examples in which an airplane mode is disabled and the user equipment **200** is restarted and powered on.
(207) For example, when the airplane mode is disabled, for the solution provided in this embodiment of the present invention, refer to FIG. **3**A and FIG. **3**B to FIG. **5**A and FIG. **5**B. The user equipment **200** shown in FIG. **3**A and FIG. **3**B is user equipment for which a SA 5G service is not provisioned, and an airplane mode is enabled on the user equipment **200**. As shown in FIG. **4**A and FIG. **4**B, after a user disables the airplane mode, the user equipment **200** displays the no-signal indicator **301**D of a mobile communication signal on the status bar. After the airplane mode is disabled on the user equipment **200**, the user equipment **200** initiates SA 5G search and registration. As shown in FIG. **5**A and FIG. **5**B, after the SA 5G registration fails, the user equipment **200** registers with the 4G network.
(208) In this embodiment of this application, after registering with the 4G network, the user equipment **200** obtains specific trigger information, to trigger the user equipment **200** to initiate SA 5G network search and registration again. As shown in FIG. **19**A and FIG. **19**B, when the user equipment **200** disconnects from the 4G network, the user equipment **200** displays the no-signal indicator **301**D of a mobile communication signal on the status bar. Then, the user equipment **200** initiates SA 5G network search and registration. As shown in FIG. **20**A and FIG. **20**B, the user equipment **200** successfully registers with the SA 5G network this time. In addition, the user equipment **200** displays, in the status bar, one or more signal strength indicators **301**H and an operator name **301**F that are of a 5G mobile communication signal, and may further display a signal strength indicator **301**G.
(209) For example, when the user equipment **200** is restarted and powered on, for the solution provided in this embodiment of the present invention, refer to FIG. **6** to FIG. **8**A and FIG. **8**B.
(210) The user equipment **200** shown in FIG. **6** is user equipment for which a SA 5G service is not provisioned. As shown in FIG. **7**A and FIG. **7**B, a user restarts the user equipment **200**. When the user equipment **200** is just powered on, the user equipment **200** displays the no-signal indicator **201**D of a mobile communication signal on the status bar **201**D. After being powered on, the user equipment **200** initiates SA 5G search and registration. As shown in FIG. **8**A and FIG. **8**B, after the

SA 5G registration fails, the user equipment **200** registers with the 4G network.
(211) In this embodiment of this application, after registering with the 4G network, the user equipment **200** obtains specific trigger information, to trigger the user equipment **200** to initiate SA 5G network search and registration again. As shown in FIG. **21**A and FIG. **21**B, when the user equipment **200** disconnects from the 4G network, the user equipment **200** displays the no-signal indicator **201**D of a mobile communication signal on the status bar **201**. Then, the user equipment **200** initiates SA 5G network search and registration. As shown in FIG. **22**A and FIG. **22**B, the user equipment **200** successfully registers with the SA 5G network this time. In addition, the user equipment **200** displays, in the status bar, one or more signal strength indicators **201**H and an operator name **201**F that are of a 5G mobile communication signal, and may further display a signal strength indicator **201**G.
(212) For ease of understanding embodiments of this application, the user equipment **200** shown in FIG. **23** is used as an example to describe user equipment to which embodiments of this application are applicable.
(213) FIG. **23** shows a schematic diagram of a structure of an example of user equipment **200** according to an embodiment of this application.
(214) The user equipment **200** may include a processor **110**, an external memory interface **120**, an internal memory **121**, a universal serial bus (universal serial bus, USB) interface **130**, a charging management module **140**, a power management module **141**, a battery **142**, an antenna **1**, an antenna **2**, a mobile communication module **150**, a wireless communication module **160**, an audio module **170**, a speaker **170**A, a receiver **170**B, a microphone **1700**, a headset jack **170**D, a sensor module **180**, a button **190**, a motor **191**, an indicator **192**, a camera **193**, a display **194**, a subscriber identification module (subscriber identification module, SIM) card interface **195**, and the like. The sensor module **180** may include a pressure sensor **180**A, a gyroscope sensor **180**B, a barometric pressure sensor **180**C, a magnetic sensor **180**D, an acceleration sensor **180**E, a distance sensor **180**E an optical proximity sensor **180**G, a fingerprint sensor **180**H, a temperature sensor **180**J, a touch sensor **180**K, an ambient light sensor **180**L, a bone conduction sensor **180**M, and the like.
(215) It may be understood that the structure illustrated in this embodiment of this application does not constitute a specific limitation on the user equipment **200**. In other embodiments of this application, the user equipment **200** may include more or fewer components than those shown in the figure, or combine some components, or split some components, or have a different component arrangement. The components shown in the figure may be implemented by hardware, software, or a combination of software and hardware.
(216) The processor **110** may include one or more processing units. For example, the processor **110** may include an application processor (application processor, AP), a modem processor, a graphics processing unit (graphics processing unit, GPU), an image signal processor (image signal processor, ISP), a controller, a memory, a video codec, a digital signal processor (digital signal processor, DSP), a baseband processor, and/or a neural-network processing unit (neural-network processing unit, NPU). Different processing units may be independent components, or may be integrated into one or more processors.
(217) The controller may be a nerve center and a command center of the user equipment **200**. The controller may generate an operation control signal based on instruction operation code and a time sequence signal, to control instruction fetching and instruction execution.
(218) The NPU may perform an artificial intelligence operation by performing convolutional neural network (convolutional neural networks, CNN) processing. For example, a CNN model is used to perform a large amount of information identification and information filtering, to implement identification for short message content.
(219) A memory may be further disposed in the processor **110**, and is configured to store instructions and data. In some embodiments, the memory in the processor **110** is a cache. The memory may store instructions or data just used or cyclically used by the processor **110**. If the

processor **110** needs to use the instructions or the data again, the processor **110** may directly invoke the instructions or the data from the memory. In this way, repeated access is avoided, waiting time of the processor **110** is reduced, and system efficiency is improved.
(220) The charging management module **140** is configured to receive a charging input from a charger.
(221) A wireless communication function of the user equipment **200** may be implemented by using the antenna **1**, the antenna **2**, the mobile communication module **150**, the wireless communication module **160**, the modem processor, the baseband processor, and the like. The antenna **1** and the antenna **2** are configured to transmit and receive electromagnetic wave signals, Each antenna in the user equipment **200** may be configured to cover one or more communication frequency bands. Different antennas may be multiplexed to improve antenna utilization.
(222) The mobile communication module **150** may provide a solution that includes wireless communication such as 2G/3G/4G/5G and that is applied to the user equipment **200**. The mobile communication module **150** may include at least one filter, a switch, a power amplifier, a low noise amplifier (low noise amplifier, LNA), and the like. The mobile communication module **150** may receive an electromagnetic wave by using the antenna **1**, perform processing such as filtering and amplification on the received electromagnetic wave, and transmit a processed electromagnetic wave to the modem processor for demodulation. The mobile communication module **150** may further amplify a signal modulated by the modem processor, and convert the signal into an electromagnetic wave for radiation through the antenna **1**. In some embodiments, at least some functional modules of the mobile communication module **150** may be disposed in the processor **110**. In some embodiments, at least some functional modules of the mobile communication module **150** and at least some modules of the processor **110** may be disposed in a same component.
(223) The modem processor may include a modulator and a demodulator. The modulator is configured to modulate a to-be-sent low-frequency baseband signal into a medium-high-frequency signal. The demodulator is configured to demodulate a received electromagnetic wave signal into a low-frequency baseband signal. Then, the demodulator transfers the low-frequency baseband signal obtained through demodulation to the baseband processor for processing. After being processed by the baseband processor, the low-frequency baseband signal is transmitted to the application processor. The application processor outputs a sound signal through an audio device (not limited to the speaker **170**A, the receiver **170**B, or the like), or displays an image or a video by using the display **194**. In some embodiments, the modem processor may be an independent component. In some other embodiments, the modem processor may be independent of the processor **110**, and is disposed in a same component as the mobile communication module **150** or another functional module.
(224) The wireless communication module **160** may provide a wireless communication solution that is applied to the user equipment **200** and that includes a wireless local area network (wireless local area networks, WLAN) (for example, a wireless fidelity (wireless fidelity, Wi-Fi) network), Bluetooth (Bluetooth, BT), a global navigation satellite system (global navigation satellite system, GNSS), frequency modulation (frequency modulation, FM), a near field, communication (near field communication, NFC) technology, an infrared (infrared. IR) technology, or the like. The wireless communication module **160** may be one or more components integrating at least one communication processing module. The wireless communication module **160** receives an electromagnetic wave through the antenna **2**, performs frequency modulation and filtering processing on the electromagnetic wave signal, and sends a processed signal to the processor **110**. The wireless communication module **160** may further receive a to-be-sent signal from the processor **110**, perform frequency modulation and amplification on the signal, and convert the signal into an electromagnetic wave for radiation through the antenna **2**.
(225) In some embodiments, in the user equipment **200**, the antenna **1** and the mobile communication module **150** are coupled, and the antenna **2** and the wireless communication

module **160** are coupled, so that the user equipment **200** can communicate with a network and another device by using a wireless communication technology. The wireless communication technology may include a global system for mobile communication (global system for mobile communications, GSM), a general packet radio service (general packet radio service, GPRS), code division multiple access (code division multiple access, CDMA), wideband code division multiple access (wideband code division multiple access, WCDMA), time-division code division multiple access (time-division code division multiple access, TD-SCDMA), long term evolution (long term evolution, LTE), BT, a GNSS, a WLAN, NFC, FM, an IR technology, and/or the like. The GNSS may include a global positioning system (global positioning system, GPS), a global navigation satellite system (global navigation satellite system, GLONASS), a BeiDou navigation satellite system (beidou navigation satellite system, BDS), a quasi-zenith satellite system (quasi-zenith satellite system, QZSS), and/or a satellite-based augmentation system (satellite based augmentation systems, SBAS).
(226) The user equipment **200** implements a display function by using the GPU, the display **194**, the application processor, and the like. The GPU is a microprocessor for image processing, and is connected to the display **194** and the application processor. The processor **110** may include one or more GPU that execute program instructions to generate or change display information.
(227) The display **194** is configured to display an image, a video, and the like. The display **194** includes a display panel. In some embodiments of this application, the display **194** displays interface content currently output by a system. For example, the interface content is an interface provided by an instant messaging application.
(228) The user equipment **200** may implement a photographing function by using the ISP, the camera **193**, the video codec, the GPU, the display **194**, the application processor, and the like. The ISP is configured to process data fed back by the camera **193**. In some embodiments, the ISP may be disposed in the camera **193**. The camera **193** is configured to capture a static image or a video.
(229) The NPU is a neural-network (neural-network, NN) computing processor. The NPU quickly processes input information by using a structure of a biological neural network, for example, by using a transfer mode between human brain neurons, and may further continuously perform self-learning. Applications such as intelligent cognition of the user equipment **100**, such as image recognition, facial recognition, speech recognition, and text understanding, may be implemented by using the NPU.
(230) The internal memory **121** may be configured to store computer executable program code, where the executable program code includes instructions. The processor **110** runs the instructions stored in the internal memory **121**, to perform various function applications and data processing of the user equipment **200**. The internal memory **121** may include a program storage area and a data storage area. The program storage area may store an operating system, an application required by at least one function (for example, a sound playing function or an image playing function), and the like. The data storage area may store data (for example, audio data, and a phone book) created during a process of using the user equipment **200**, and the like.
(231) The user equipment **200** may implement audio functions by using the audio module **170**, the speaker **170**A, the receiver **170**B, the microphone **170**C, the headset jack **170**D, the application processor, and the like. For example, a music playback function and a recording function are implemented. The audio module **170** is configured to convert digital audio information into an analog audio signal output, and is also configured to convert an analog audio input into a digital audio signal. The speaker **170**A, also referred to as a “loudspeaker”, is configured to convert an audio electrical signal into a sound signal. The receiver **170**B, also referred to as an “earpiece”, is configured to convert an audio electrical signal into a sound signal. The microphone **170**C, also referred to as a “mike” or a “microphone”, is configured to convert a sound signal into an electrical signal. The headset jack **170**D is configured to connect to a wired headset.
(232) The pressure sensor **180**A is configured to sense a pressure signal, and may convert the

pressure signal into an electrical signal. The gyroscope sensor **180**B may be configured to determine a motion posture of the user equipment **200**. The barometric pressure sensor **180**C is configured to measure barometric pressure. The magnetic sensor **180**D includes a Hall sensor.

(233) The user equipment **200** may detect opening and closing of a flip leather case by using the magnetic sensor **180**D. The acceleration sensor **180**E may detect magnitudes of accelerations in various directions generally on three axes) of the user equipment **200**. The distance sensor **180**F is configured to measure a distance. The optical proximity sensor **180**G may include a light-emitting diode (LED) and an optical detector, for example, a photodiode. The ambient light sensor **180**L is configured to sense ambient light brightness. The fingerprint sensor **180**H is configured to collect a fingerprint. The temperature sensor **180**J is configured to detect a temperature. The touch sensor **180**K is also referred to as a "touch panel". The touch sensor **180**K may be disposed on the display **194**. The touch sensor **180**K and the display **194** constitute a touchscreen, and the touchscreen is also referred to as a "touch control screen". The touch sensor **180**K is configured to detect a touch operation performed on or near the touch sensor **180**K The touch operation is an operation that a user touches the display **194** by using a hand, an elbow, a stylus, or the like. The touch sensor may transfer the detected touch operation to the application processor, to determine a type of a touch event. The bone conduction sensor **180**M may obtain a vibration signal.

(234) The button **190** includes a power button, a volume button, and the like. The button **190** may be a mechanical button, or may be a touch button. The motor **191** may generate a vibration prompt. The indicator **192** may be an indicator light, and may be configured to indicate a charging status and a power change, or may be configured to indicate a message, a missed call, a notification, and the like.

(235) A software system of the user equipment **200** may use a layered architecture, an event-driven architecture, a microkernel architecture, a microservice architecture, or a cloud architecture. In this embodiment of this application, an Android system with a layered architecture is used as an example to describe a software structure of the user equipment **200**.

(236) FIG. **24** is a block diagram of a software structure of the user equipment **200** according to an embodiment of this application.

(237) In the layered architecture, software is divided into several layers, and each layer has a clear role and a clear task. The layers communicate with each other by using a software interface. In some embodiments, the Android system is divided into four layers: an application layer, an application framework layer, an Android Runtime (Android runtime) and system library, and a kernel layer from top to bottom.

(238) The application layer may include a series of application packages.

(239) As shown in FIG. **24**, the application packages may include applications such as Camera, Gallery, Calendar, Call, Maps, Navigation, WLAN, Bluetooth, Music, Videos, and Messages.

(240) In this application, a floating launcher (floating launcher) may be further added to the application layer, to serve as a default display application in the foregoing floating window, and provide an entry for a user to enter another application.

(241) The application framework layer provides an application programming interface (application programming interface, API) and a programming framework for an application at the application layer. The application framework layer includes some predefined functions.

(242) As shown in FIG. **24**, the application framework layer may include a window manager (window manager), a content provider, a view system, a phone manager, a resource manager, a notification manager, an activity manager (activity manager), and the like.

(243) The window manager is configured to manage a window program. The window manager may obtain a size of a display, determine Whether there is a status bar, lock the display, take a screenshot of the display, and the like. In this application, a FloatingWindow may be extended based on a native PhoneWindow of Android, and is specially configured to display the foregoing floating window, to distinguish the floating window from a common window. The window has an

attribute of being displayed on a top layer of a series of windows in a floating manner. In some optional embodiments, a proper value of a size of the window may be given based on an actual screen size according to an optimal display algorithm. In some possible embodiments, an aspect ratio of the window may be an aspect ratio of a screen of a conventional mainstream mobile phone by default. In addition, to help the user close and hide the floating window, an extra close button and an extra minimize button may be drawn in an upper right corner.

(244) The content provider is configured to: store and obtain data, and enable the data to be accessed by an application. The data may include a video, an image, audio, calls that are made and received, a browsing history and bookmarks, a phone book, and the like.

(245) The view system includes a visual control, for example, a control for displaying text or a control for displaying a picture. The view system may be used to build an application. A display interface may include one or more views. For example, a display interface including a short message notification icon may include a text display view and a picture display view. In this application, a key view used for operations such as closing and minimization may be correspondingly added to the floating window, and bound to the FloatingWindow in the window manager.

(246) The phone manager is configured to provide a communication function of the user equipment **200**, for example, management of a call status (including answering, declining, or the like).

(247) The resource manager provides various resources for an application, such as a localized character string, an icon, a picture, a layout file, and a video file.

(248) The notification manager enables an application to display notification information in a status bar **207**, and may be configured to convey a notification-type message. The displayed information may automatically disappear after a short pause without user interaction. For example, the notification manager is configured to: notify download completion, give a message notification, and the like. The notification manager may alternatively be a notification that appears in a top status bar of the system in a form of a graph or a scroll bar text, for example, a notification of an application running in the background, or may be a notification that appears on a display in a form of a dialog window. For example, text information is prompted in the status bar, a prompt tone is played, the user equipment vibrates, or an indicator light blinks.

(249) The activity manager is configured to manage activities that are running in the system, including a process (process), an application, a service (service), task (task) information, and the like. In this application, an activity task stack dedicated to managing an activity of an application displayed in the floating window may be newly added to an activity manager module, to ensure that the activity and a task of the application in the floating window do not conflict with an application displayed on the screen in full screen.

(250) In this application, a motion detector (motion detector) may be further added to the application framework layer, to perform logical determining on an obtained input event and identify a type of the input event. For example, the motion detector determines, based on information such as touch coordinates and a timestamp of a touch operation included in the input event, that the input event is a knuckle touch event, a finger pad touch event, or the like. In addition, the motion detector may further record a track of the input event, determine a gesture rule of the input event, and respond to different operations based on different gestures.

(251) The Android Runtime includes a kernel library and a virtual machine. The Android runtime is responsible for scheduling and management of the Android system.

(252) The kernel library includes two parts: a function that needs to be invoked in Java language and a kernel library of Android.

(253) The application layer and the application framework layer run on the virtual machine. The virtual machine executes Java files at the application layer and the application framework layer as binary files. The virtual machine is configured to implement functions such as object lifecycle management, stack management, thread management, security and exception management, and

garbage collection.
(254) The system library may include a plurality of function modules, for example, an input manager (input manager), an input dispatcher (input dispatcher), a surface manager (surface manager), a media library (Media Libraries), a three-dimensional graphics processing library (for example, OpenGL ES), and a 2D graphics engine (for example, SGL).
(255) The input manager is responsible for obtaining event data front an underlying input driver, parsing and encapsulating the event data, and sending the event data to the input dispatcher.
(256) The input dispatcher is configured to store window information. After receiving an input event from the input manager, the input dispatcher searches for a proper window in windows stored in the input dispatcher and dispatches the event to the window.
(257) The surface manager is configured to manage a display subsystem, and provide fusion of 2D and 3D layers for a plurality of applications.
(258) The media library supports playback and recording in a plurality of commonly used audio and video formats, static image files, and the like. The media library may support a plurality of audio and video coding formats, for example, MPEG 4, H.264, MP3, AAC, AMR, JPG, and PNG.
(259) The three-dimensional graphics processing library is configured to implement three-dimensional graphics drawing, image rendering, composition, layer processing, and the like.
(260) The 2D graphics engine is a drawing engine for 2D drawing.
(261) The kernel layer is a layer between hardware and software. The kernel layer includes at least a display driver, a camera driver, an audio driver, and a sensor driver.
(262) For ease of understanding embodiments of this application, a network device **300** shown in FIG. **25** is used as an example to describe a network device to which embodiments of this application are applicable.
(263) FIG. **25** shows the network device **300** according to an embodiment of this application. As shown in FIG. **25**, the network device **300** may include one or more network device processors **310**, a memory **320**, and a communication interface **330**. These components may be connected by using a bus **340** or in another manner. In FIG. **25**, an example in which these components are connected by using a bus is used.
(264) The communication interface **330** may be used by the network device **300** to communicate with another communication device, for example, user equipment or another network device. Specifically, the communication interface **330** may be a 5G communication interface, or may be a future new radio communication interface. In addition to a wireless communication interface, the network device **300** may be configured with a wired communication interface **330** to support wired communication. For example, a backhaul link between a network device **300** and another network device **300** may be a wired communication connection.
(265) The memory **320** is coupled to the network device processor **310**, and is configured to store various software programs and/or a plurality of sets of instructions. Specifically, the memory **320** may include a high-speed random access memory, and may further include a non-volatile memory, for example, one or more magnetic disk storage devices, a flash memory device, or another non-volatile solid-state storage device. The memory **320** may store an operating system (briefly referred to as a system below), for example, an embedded operating system such as uCOS, VxWorks, or RTLinux. The memory **320** may further store a network communication program. The network communication program may be used to communicate with one or more additional devices, one or more pieces of user equipment, one or more network devices, or the like.
(266) In this embodiment of this application, the network device processor **310** may be configured to read and execute computer-readable instructions. Specifically, the network device processor **310** may be configured to: invoke a program stored in the memory **32**, for example, a program for implementing, on a network device **300** side, a network slice management method provided in one or more embodiments of this application; and execute instructions included in the program.
(267) It may be understood that the network device **300** may be the AN **102** in the communication

system **100** shown in FIG. **1**, or may be the core network entity in the communication system **100** shown in FIG. **1**, for example, a UPF **104**, an AMF **105**, an SMF **106**, an AUSF **107**, an NSSF **108**, an NEF **109**, an NRF **110**, a PCF **111**, or a UDM **112**.

(268) It should be noted that the network device **300** shown in FIG. **25** is merely an implementation of this embodiment of this application. In actual application, the network device **300** may further include more or fewer components. This is not limited herein.

(269) An embodiment of this application further provides a chip system **400**. The chip system **400** may include one or more processors **410**.

(270) The processor **410** may be an integrated circuit chip and has a signal processing capability. In an implementation process, the steps in the foregoing methods may be completed by using a hardware integrated logic circuit in the processor **410**, or by using instructions in a form of software. The processor **410** may include a modem processor, and may further include an application processor (application processor, AP). The processor **410** may implement or perform the methods and the steps disclosed in embodiments of this application. The processor **410** may further include more components. This is not limited herein.

(271) Optionally, the chip system **400** further includes a memory **420**. The memory **420** may include a read-only memory and a random access memory, and provide operation instructions and data to the processor. A part of the memory **420** may further include a non-volatile random access memory.

(272) Optionally, the memory **420** stores an executable software module or a data structure, and the processor **420** may perform a corresponding operation by invoking operation instructions (the operation instructions may be stored in an operating system) stored in the memory.

(273) Optionally, the chip system **400** may be disposed in the user equipment in this embodiment of this application. Optionally, the processor **410** is configured to perform the processing steps of the user equipment in the embodiments shown in FIG. **9**, FIG. **10**A and FIG. **10**B, and FIG. **15**A and FIG. **15**B to FIG. **18**A and FIG. **18**B. For example, the chip system **400** sends a registration request by using the modem processor, and the chip system **400** parses a notification short message by using the AP processor. The memory **420** is configured to store data and instructions of the user equipment in the embodiments shown in FIG. **9**, FIG. **10**A and FIG. **10**B, and FIG. **15**A and FIG. **15**B to FIG. **18**A and FIG. **18**B.

(274) An embodiment of this application further provides a computer-readable storage medium. All or some of the methods described in the foregoing method embodiments may be implemented by using software, hardware, firmware, or any combination thereof. When the methods are implemented in the software, functions used as one or more instructions or code may be stored in the computer-readable medium or transmitted on the computer-readable medium. The computer-readable medium may include a computer storage medium and a communication medium, and may further include any medium that can transfer a computer program from one place to another. The storage medium may be any available medium accessible to a computer.

(275) An embodiment of this application further provides a computer program product. All or some of the methods described in the foregoing method embodiments may be implemented by using software, hardware, firmware, or any combination thereof. When the methods are implemented in the software, all or some of the methods may be implemented in a form of the computer program product. The computer program product includes one or more computer instructions. When the computer program instructions are loaded and executed on a computer, all or some of the procedures or functions that are described in the foregoing method embodiments are generated. The computer may be a general-purpose computer, a dedicated computer, a computer network, a network device, user equipment, or another programmable apparatus.

(276) In conclusion, the foregoing embodiments are merely intended for describing the technical solutions of this application, rather than limiting this application. Although this application is described in detail with reference to the foregoing embodiments, a person of ordinary skill in the

art should understand that modifications may still be made to the technical solutions described in the foregoing embodiments, or equivalent replacements may still be made to some technical features thereof. These modifications or replacements do not make the essence of the corresponding technical solutions depart from the scope of the technical solutions of embodiments of this application.

Claims

1. A network search method for standalone (SA) fifth generation (5G) service auto-provisioning implemented by a user equipment, wherein the network search method comprises: sending, to a network side device, a registration request requesting to register the user equipment with a 5G network, wherein a networking mode of the 5G network is SA networking; receiving, from the network side device, a registration reject response when no available 5G subscription data corresponding to the user equipment is obtained through a query, wherein the registration reject response notifies the user equipment that a registration with the 5G network has failed; receiving, from the network side device after the network side device provisions a 5G service for the user equipment, a notification short message notifying the user equipment that the 5G service is successfully provisioned; and resending, in response to the notification short message and to the network side device, the registration request.
2. The network search method of claim 1, further comprising further sending, to the network side device for a first time, the registration request.
3. The network search method of claim 1, further comprising: identifying that the user equipment has no network service in a New Radio (NR) cell; and further sending, in response to identifying that the user equipment has no network service in the NR cell and to the network side device for a first time, the registration request.
4. The network search method of claim 1, further comprising: leaving a Long-Term Evolution (LTE) cell; and entering a New Radio (NR) cell; and further sending, in response to leaving the LTE cell and entering the NR cell and to the network side device for a first time, the registration request.
5. The network search method of claim 1, further comprising: identifying that the user equipment is powered on in a New Radio (NR) cell; and further sending, in response to identifying that the user equipment is powered on in the NR cell and to the network side device for a first time, the registration request.
6. The network search method of claim 1, further comprising: identifying that an airplane mode of the user equipment is disabled in a New Radio (NR) cell; and further sending, in response to identifying that the airplane mode of the user equipment is disabled in the NR cell and to the network side device for a first time, the registration request.
7. The network search method of claim 1, wherein the user equipment supports the 5G network.
8. The network search method of claim 1, wherein after resending the registration request to the network side device, the network search method further comprises receiving, a from the network side device after the available 5G subscription data is obtained through the query, a registration accept response notifying the user equipment that the registration with the 5G network has succeeded.
9. The network search method of claim 1, further comprising: detecting that the notification short message comprises a preset keyword; and further resending, in response to detecting that the notification short message comprises the preset keyword and to the network side device, the registration request.
10. A user equipment comprising: a memory configured to store instructions; and a processor coupled to the memory and configured to execute the instructions to cause the user equipment to: send, to a network side device, a registration request requesting to register the user equipment with

a fifth generation (5G) network, wherein a networking mode of the 5G network is standalone (SA) networking; receive, from the network side device when no available 5G subscription data corresponding to the user equipment is obtained through a query, a registration reject response notifying the user equipment that a registration with the 5G network has failed; receive, from the network side device after the network side device provisions a 5G service for the user equipment, a notification short message notifying the user equipment that the 5G service is successfully provisioned; and resend, to the network side device in response to the notification short message, the registration request.

11. The user equipment of claim 10, wherein the processor is further configured to execute the instructions to cause the user equipment to further send the registration request to the network side device for a first time.

12. The user equipment of claim 10, wherein the processor is further configured to execute the instructions to cause the user equipment to: identify that the user equipment has no network service in a New Radio (NR) cell; and further send, in response to identifying that the user equipment has no network service in the NR cell and to the network side device for a first time, the registration request.

13. The user equipment of claim 10, wherein the processor is further configured to execute the instructions to cause the user equipment to: leave a Long-Term Evolution (LTE) cell and enter a New Radio (NR) cell; and further send, in response to leaving the LTE cell and entering the NR cell and to the network side device for a first time, the registration request.

14. The user equipment of claim 10, wherein the processor is further configured to execute the instructions to cause the user equipment to: identify that the user equipment is powered on in a New Radio (NR) cell; and, further send, in response to identifying that the user equipment is powered on in the NR cell and to the network side device for a first time, the registration request.

15. The user equipment of claim 10, wherein the processor is further configured to execute the instructions to cause the user equipment to: identify that an airplane mode of the user equipment is disabled in a New Radio (NR) cell; and further send, in response to identifying that the airplane mode of the user equipment is disabled in the NR cell and to the network side device for a first time, the registration request.

16. The user equipment of claim 10, wherein the user equipment supports the 5G network.

17. The user equipment of claim 10, wherein after resending the registration request to the network side device, the processor is further configured to execute the instructions to cause the user equipment to receive, from the network side device after the available 5G subscription data is obtained through the query, a registration accept response notifying the user equipment that the registration with the 5G network has succeeded.